

EUROPEAN COMMISSION

DIRECTORATE-GENERAL FOR COMMUNICATIONS NETWORKS, CONTENT ANDTECHNOLOGY

Digital Excellence & Science Infrastructure

Technologies for Smart Communities

D05.04 Public report on the roadmap for the scale-up of EU AI-powered DT

CNECT/2022/OP/0098 – Procurement of the Technical Specifications for the Local Digital Twins (LDTs) Toolbox

This study has been prepared as part of the Local Digital Twins Toolbox project. The project is an initiative of the European Commission. Directorate-General for Communications Networks, Content and Technology of the European Union is responsible for contract management of the Local Digital Twins Toolbox project.

European Commission
Directorate-General for Communications Networks, Content and Technology
Unit C.3 Technologies for Smart Communities

Last update: 14 August 2023

Task	TASK-05: Outline design of the LDT Toolbox and draft procurement specifications
Deliverable	D05.04: Public report on the roadmap for the scale-up of EU AI-powered DT
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ISBN: 978-92-68-06529-7 doi: 10.2759/345126 KK-09-23-370-EN-N

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Abbreviations and Acronyms

List of Abbreviations and Acronyms	
AI	Artificial Intelligence
AR	Augmented Reality
API	Application Programming Interface
BBs	Building Blocks
DEP	Digital Europe Programme
DIGISER	Digital Innovation in Governance and Public Service Provision
EC	European Commission
EDIC	European Digital Infrastructure Consortium
EU	European Union
ICT	Information and Communications Technology
IoT	Internet-of-Things
ITU	International Telecommunication Union
KPIs	Key Performance Indicators
LDT	Local Digital Twin
MIMs	Minimal Interoperability Mechanisms
OASC	Open & Agile Smart Cities
SCC(s)	Smart Cities and Communities
TEFs	Testing and Experimentation Facilities

1. Introduction

1.1 Context

The European Union (EU)'s strategic focus on **advancing the digital transformation of smart cities and communities** giving rise to a wide ecosystem of initiatives, not only contributing to Europe's digital transition but also intertwined with the green transition under the European Green Deal umbrella.

The major initiatives underpinning this ecosystem include the Digital Europe Programme (DEP), the Living-in.eu movement, and common European data spaces. The European Commission's (EC) communication "2030 Digital Compass: The European way for the Digital Decade", places the DEP as a comprehensive EU funding programme to reinforce the EU's critical digital capacities in key areas of strategic importance, including the wide adoption of digital technologies across the economy and society. In addition, through the "Living-in.eu Declaration", representatives of public administrations at local, regional, national, or European level have committed to a set of principles linked to citizen-centricity, city-led approach, open innovation, ethical use of data, use of technology, and interoperability[1].

In particular, the EC is in the process of deploying several major Artificial Intelligence (AI)-oriented initiatives that seek to dramatically advance the ability of communities within Europe to exploit digital technologies to enhance the lives of their citizens and to tackle the climate crisis. The vision is specifically to:

- Enable the implementation of AI enabled Local Digital Twins (LDTs) to greatly improve the speed and effectiveness of planning and enable management of the local community to be based on accurate and near real time data.
- Develop and implement a Citizen-focused version of the metaverse – the CitiVerse, that will utilise augmented reality applications, built on the LDTs, to provide greater support to the citizen in playing an active and informed role in the management of their community.
- Support sectoral and community-wide data spaces to make the data being generated within the community available to share and re-use in appropriate ways to support the citizen and enable effective action to address the climate crisis.

In this project, an **LDT** is defined as a "virtual representation of the physical assets, processes, and people within a geographically located community, which reflect and derive from cross-sectorial, historical and (near) real-time data. Its purpose is to enhance evidence-based decision-making, at the operational, strategic, and tactical levels, to better meet the needs of communities. LDTs combine multiple technologies, such as data analytics and artificial intelligence, enabling predictive and simulation models that can be updated and changed as their physical equivalents change". In line with this, in this project, "**local**" refers to communities, i.e., geographically, or territorially defined regions, peri-urban areas, cities, city districts, towns, rural areas, and villages as well, where cities and communities' administrations play a key role.

ESPON's Digital Innovation in Governance and Public Service Provision (DIGISER) project, which surveyed European cities in 2021/22 found that only 20% of European cities surveyed make use of interoperable digital solutions or services (namely ISA2, CEF Digital Building Blocks, Minimal Interoperability Mechanisms (MIMs), FIWARE) and only 35% of public authorities operate or work with urban data platforms. However, it also found that out of those cities with an urban data platform in place, almost 60% have already integrated or are planning to integrate their data platforms with an LDT or similar.

[1] For more details on the principles of the Living-in.eu Declaration please consult: Declaration on joining forces to boost sustainable digital transformation in cities and communities in the EU | Living in EU (living-in.eu)

LDTs can be at various levels of complexity and value. The analysis of the current state of LDT solutions implemented or under development by cities and communities in Europe is based on the **DUET Digital Twin Maturity Model**. This model relies on four classification criteria: Level 1 – Awareness of Twins, Level 2 – Experimental Twins, Level 3 – Predictive Twins, Level 4 – Intelligent Twins. These criteria are employed to define four phases of maturity: the Strategy Phase, the Exploratory Phase, the Insightful Phase, and the Future Ready Phase. Each level of maturity from Awareness of Twins to Intelligent Twins underscores the increasing integration and influence of AI on the Digital Twins' capabilities. Figure 1 illustrates the classification criteria related to the four maturity levels.

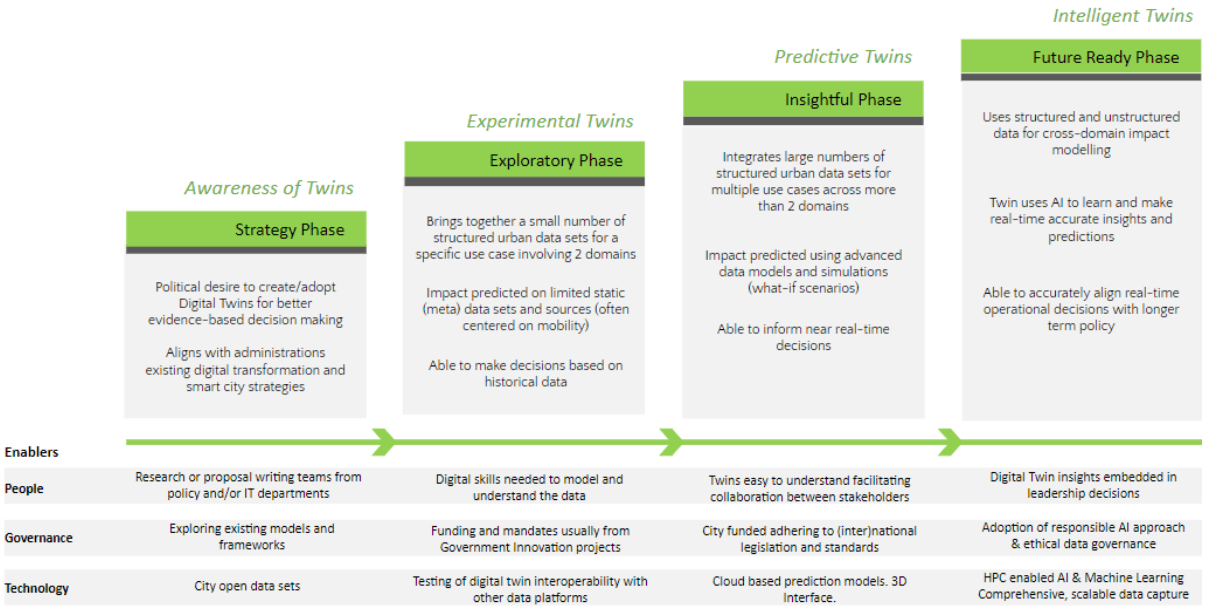


Figure 1: DUET Digital Twin Maturity Model

Source: Digital Twin Maturity Model | DUET (digitalurbantwins.com)

In this context, the DEP funds important initiatives to support EU cities and communities characterised by different stages of digital maturity and LDT ambition levels. One upcoming DEP-funded project on "Advancing initial stages for the transformation of smart communities"[2] will aim to provide the tools and support required to increase the awareness and readiness of less advanced communities. In addition to this, the current project, "CNECT/2022/OP/0098 - Procurement of the Technical Specifications for the Local Digital Twins (LDTs) Toolbox", brings forward a proposal for an **AI-driven EU LDT Toolbox** that will support more advanced EU communities in their digital transformation journey. This project proposes that the LDT Toolbox takes the form of a "one-stop-shop" platform where a selection of building blocks (BBs) will be made available to cities and communities to support the development of their LDTs. The concept of a 'building block' provides a level of abstraction that translates the desired capabilities into specifications for possible 'tools' that fulfil the needs of an LDT, in a technology-agnostic and vendor-agnostic way to ensure interoperability. Each BB can be considered as an accelerator for the implementation and operationalisation of LDTs[3]. Furthermore, the LDT Toolbox itself will be developed and deployed in a following project, "Development of the Local Digital Twins (LDT) Toolbox for Advancing the Transformation of Smart Communities".

[2] See previous call for tenders https://etendering.ted.europa.eu/cft/cft-display.html?cftId=11876

[3] The public report "D05.02 LDT Toolbox detailed specifications" provides more detailed information on the prioritised BBs of the LDT Toolbox and their respective technical specifications.

1.2 Purpose & Scope

This document is a formal deliverable of the "CNECT/2022/OP/0098 - Procurement of the Technical Specifications for the Local Digital Twins (LDTs) Toolbox" project. The purpose of this deliverable is to bring forward a proposal for a **high-level roadmap for the LDT Toolbox's geographical deployment, technical scale-up, and evolution** considering:

- the complexity of the wider ecosystem of initiatives[4] aiming to advance the digital transformation of smart cities and communities, and the deployment of AI enabled Digital Twins; and
- the existing disparities of European cities and communities both in terms of overall digital maturity, and regarding the ambition levels of their LDTs.

In particular, an LDT Toolbox's **geographical deployment** will not only be vertical from different local scales (e.g., from cities to peri-urban areas) but also horizontally from geographically concentrated and more advanced/digitally mature countries to peripheral European countries with considerable market growth potential. Moreover, an LDT Toolbox deployment will leverage on the ecosystem initiatives, as it aims to foster interoperability while leveraging synergies with existing technologies, such as data platforms and AI-powered Digital Twins developed through other EU-funded initiatives.

The **scale-up** of an LDT Toolbox can, in some senses, be considered as technology-driven, given that it aims to meet the needs of EU cities and communities characterised by all levels of maturity from level 0 to level 4. More importantly, the scale-up of an LDT Toolbox will be **content and community-driven** as it will evolve in a way that stays close to the users and is able to respond to their policy and service provision priorities. Figure 2 illustrates the LDT Toolbox's geographical deployment as well as its scale-up and evolution in the wider ecosystem of initiatives.

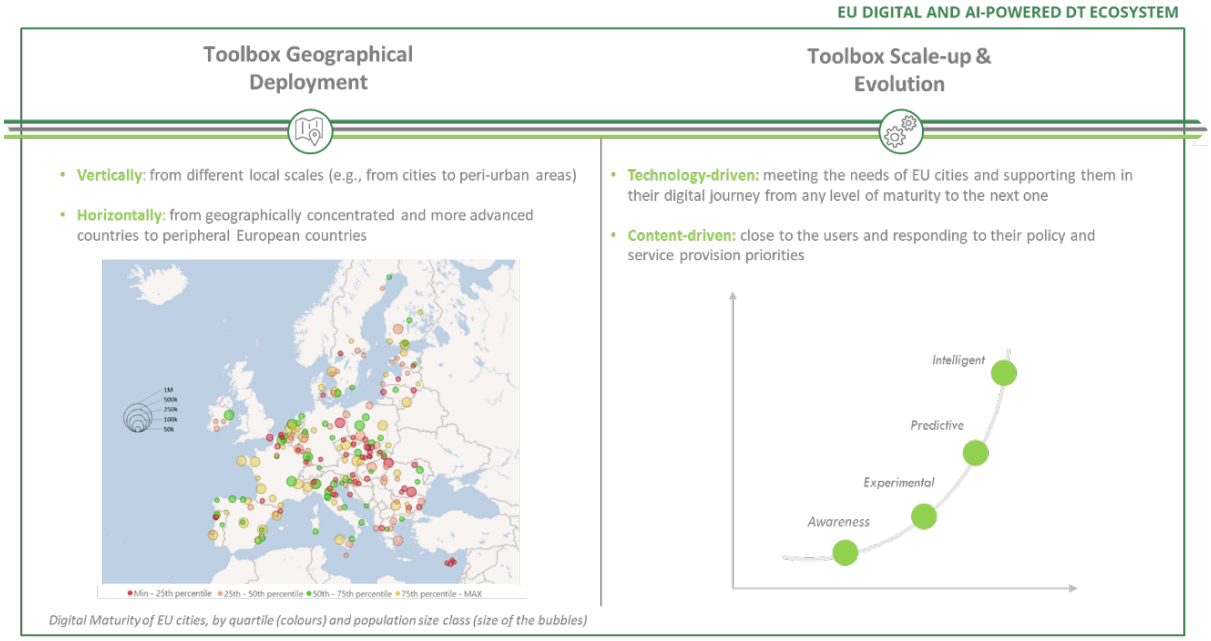


Figure 2: Toolbox deployment, scale-up and evolution in the wider EU digital and AI-powered Digital Twin ecosystem

Source: Consortium

In concrete terms, the high-level roadmap proposed in this document will highlight how an LDT Toolbox can contribute to the journey EU cities and communities need to follow and the wider deployment of AI-enabled LDTs. It will consider the situation of EU cities and communities at the moment with respect to LDTs, the challenges they face and the support that they need. The

[4] See Annex 1 (section 6.1) for a first listing of these initiatives.

coordination and alignment with national and regional priorities and initiatives will also assume particular relevance to maximise the potential for pooling resources, testing, and scaling solutions locally. As the EU digital and AI-powered Digital Twins ecosystem is fast-evolving, aspects such as technology updates, sustainability and maintenance over time will need to be duly captured in the roadmap.

1.3 Approach & Structure

To produce this deliverable, the project team analysed the EU digital ecosystem and the wider development of AI enabled Digital Twins given that an LDT Toolbox will contribute to this ecosystem and leverage synergies with existing technologies and solutions developed through other EU-funded initiatives. The project team has gathered all crucial inputs for the action definition and timing of the roadmap for the geographical deployment, technical scale-up, and evolution of the LDT Toolbox in the wider EU digital and AI-powered Digital Twins' ecosystem. All sections presented below include a sub-section summarising the key inputs for the roadmap as well a visual highlighting the related roadmaps actions.

The roadmap itself is composed of **four streams** representing the multiple co-dependent activities which should occur simultaneously over the short-, mid-, and long-term so that an LDT Toolbox's geographical deployment, technological scale-up, and evolution can be positioned within the wider deployment of AI-enabled LDTs, and ecosystem of EU-funded initiatives. The first stream is the **AI-powered ecosystem development** which provides an overview of the various activities and projects that will affect the EU digital and AI-powered Digital Twin ecosystem as a whole. The second stream is the **LDT Toolbox deployment, scale-up and evolution** which include the activities mainly supporting the LDT Toolbox's geographical deployment (vertically and horizontally). The **Awareness increase** is the third stream which refers to all the activities which will need to occur to raise awareness and interest in the Toolbox. **The Community Engagement** is the fourth and final stream which include the activities required to engage with EU cities and communities not only to regularly share best practices in the development of AI-powered LDTs but also to gather user feedback and opinions on the LDT Toolbox. The actions included in the streams are enriched by sections 2, 3 as well as 4 of this report.

Section 2 describes the EU digital ecosystem and the wider development of AI enabled LDTs, in which an LDT Toolbox will be deployed and scaled-up. It sheds light on the high-level contextual timeline of the wider deployment of AI enabled LDTs based on the EC's next tender phases. It also provides a first mapping of the diverse initiatives that lay the foundation for LDTs so as to identify different types of synergies that an LDT Toolbox's geographical deployment and technical scale-up will need to consider. In addition, it emphasises the lessons learnt from front-runner cases on their journey towards higher ambitions and maturity in the development of data platforms and LDTs over time as well as the main challenges Smart Cities and Communities (SCCs) face.

Section 3 maps the main European organisations active in community engagement which will need to be involved in the geographical deployment as well the technical scaling-up and evolution of the LDT Toolbox. In addition, it describes the main community engagement actions associated with the three phases of the high-level contextual timeline presented in Section 2.1.

Section 4 presents the business sustainability plan of an LDT Toolbox derived from the following core sustainability principles: strategic sustainability, stakeholders ecosystem sustainability, financial sustainability, environmental sustainability, technological sustainability and legal sustainability. In addition, it introduces a value model for the maintenance and evolution of an LDT Toolbox based on a strong balance between ensuring the high availability of BBs and tools (BBs Utilisation) and fostering reduced maintenance costs (cost control).

Finally, **Section 5 consolidates the key inputs of the previous sections and describes the major milestones of the roadmap** for the geographical deployment, technical scale-up, and evolution of the LDT Toolbox in the wider EU digital and AI-powered Digital Twins ecosystem based on guiding

principles and assumptions. In addition, it provides an overview of risks and mitigation actions associated with the scale-up and evolution of an LDT Toolbox as well as a set of Key Performance Indicators (KPIs) for the assessment and progress monitoring.

2. EU digital and AI-powered Digital Twin ecosystem

To maximise the benefits of an LDT Toolbox, the roadmap is developed based on the wider efforts of the EC for enabling and supporting the EU digital transformation. First, the main phases of the roadmap are aligned with the upcoming DEP-funded project phases (2.1). Second, the actions of the Toolbox will be aligned with the actions of EU initiatives – from data platforms to AI-powered Digital Twins (2.2). Third, lessons learned from more advanced EU city platforms are used to explore the idea of a City Guide on AI-powered Digital Twins (2.3). Last but not least, SCC key challenges are analysed to feed the first strategic actions of the roadmap (2.4).

2.1 DEP-funded project phases

The wider deployment of AI-enabled LDTs is based on actions and developments out of the scope of the present project. However, an educated assumption can be made for a high-level timeline, considering the other tenders in an LDT set of procurements[5]. They will provide the main context for the roll out of LDTs and of an LDT Toolbox. The timeline for the deployment of AI enabled LDTs can be seen as having three phases; Short-, medium-, and long-term.

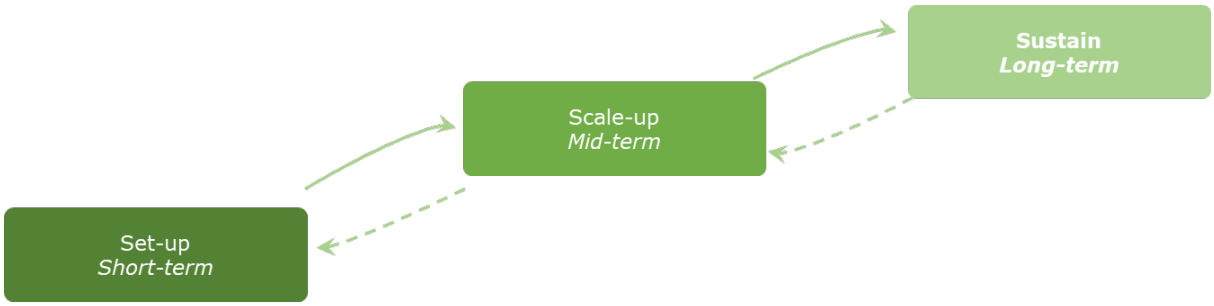


Figure 3: Project phases

Source: Consortium

Short term – Set-up

In September 2023, the EC announced the two-year project "Local Digital Twins Toolbox procurement - Advancing initial stages for the transformation of Smart Communities". This project will be focused on supporting communities at the very beginning of their digital transformation journey, laying the groundwork for an LDT Toolbox and preparing for its wide-scale adoption across the EU.

Following the call for tenders to be published in the beginning of September entitled "Development of the Local Digital Twins (LDT) Toolbox for Advancing the Transformation of Smart Communities"[6], the project is likely to kick-off six months after. The objective is to develop a Toolbox of key open-source tools to support local communities in implementing LDTs, along with a process for the ongoing development of those tools, as well as the monitoring and evaluation. It is divided in two lots:

- Lot 1: development of the EU Local Digital Twins (EU LDT) Toolbox for advancing the transformation of Smart Communities,
- Lot 2: monitoring, evaluation, and quality review of the development of the EU Local Digital Twins (EU LDT) Toolbox for Advancing the transformation of Smart Communities.

[5] Funding | Shaping Europe's digital future (europa.eu)

[6] https://ted.europa.eu/udl?uri=TED:NOTICE:438605-2023:TEXT:EN:HTML&src=0

During the first year of work of an LDT Toolbox development and deployment, the Community Organisation and the Tool Development Task Force[7] proposed for the LDT Toolbox will be set up and will start working, with already a significant number of useful tools incorporated within the Toolbox by the end of this phase[8].

Another significant step towards LDT deployment across Europe is the "Networked Local Digital Twins towards the CitiVerse" European Digital Infrastructure Consortium (EDIC) [9]. This was announced at the first Living-in.EU Mayors Digital Assembly in June 2023. The EDIC is an instrument of the European Commission's Digital Decade Policy Programme 2030 and creates a vehicle for shared investments in knowledge and funding at European scale. It was initially led by Estonia with support from Germany, Slovenia, Czech Republic, and Spain, and has already attracted further member states to join. This initiative aims to facilitate the deployment of digital tools in Europe. The focus starts with LDTs and is gradually moving towards the development of the CitiVerse – a citizen-focused metaverse.

The EDIC will take some time to be set up – the first working meeting is foreseen in November 2023. It is likely that work on developing the structure and priorities and gaining the necessary agreements with the Member States concerned will take around 18 months so it will not play a direct role in this short term phase, as it will likely not be ready for launch until early 2025. However, given that the intention is that the EDIC will take over the long-term oversight of the Toolbox, during this phase there will need to be close liaison between the officials designing the EDIC and those developing the Toolbox governance structures, to enable a smooth handover process later.

This short-term phase will capitalise on the work that is already well underway to develop resources for communities around Net-zero, data spaces and AI. A significant initiative in this area is the AI4EU[10] project, an initiative to foster development and adoption of AI solutions in various sectors across Europe. By providing resources, tools, and datasets for AI researchers and practitioners, AI4EU is positioned to play a pivotal role in the EU's AI strategy. Section 2.2 will give a high-level overview of existing initiatives.

So, by the end of the first year of the Toolbox Development and Deployment Project, not only should there be a good number of tools ready for use, with processes having been tried and tested, the wider support coming from the other initiatives will mean that everything is ready for scaling up the deployment of the Toolbox within communities throughout the EU.

[7] A detailed explanation can be found in the 2.1 & 2.2 sections of the report D05.03: "Public report on the roadmap proposal for the deployment of the LDT Toolbox"

[8] See the report 05.03 "Roadmap proposal for the deployment of the LDT Toolbox"

[9] An EDIC , is an instrument of the Commission Digital Decade Policy Programme 2030 and creates a vehicle for shared investments in knowledge and funding at European scale. More information here : https://living-in.eu/news/launch-european-funding-instrument-upscale-digital-twins-towards-citiverse-through-living-ineu ; and here: https://digital-strategy.ec.europa.eu/en/news/digital-assembly-2023-digital-open-and-secure-europe

[10] https://www.ai4europe.eu/

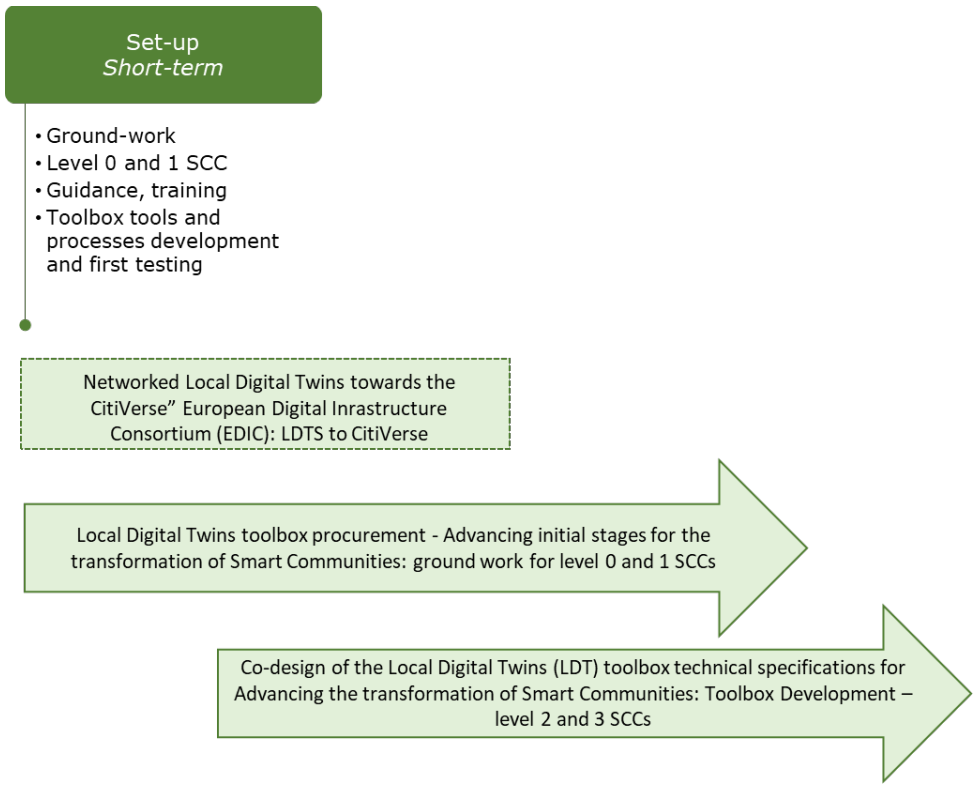


Figure 4: Short term set-up

Source: Consortium

Medium term – Scale-up

This will involve a high-profile launch, with the aim of fast growth, both in the number of communities using the Toolbox and in the number of tools being developed. It will also be the start of engaging with those communities at the early stages of digital transformation in helping them begin to plan their roadmap for LDT implementation.

By around Spring 2025 it is expected that:

- The support services provided by the "Advancing initial stages for the transformation of Smart Communities" project would have been delivering value for enough time to gain a solid evaluation of the value they are providing. This would be a good time to review their longer-term future and to have a major push to scale up usage.
- The Community Organisation should have gained many members from communities, industry, academia and civic society and be demonstrating value to its members. This would be a good time for a new push to gain new members and to raise its profile.
- The Tool Development Task Force, the open-source community that will be increasingly taking on the role of developing and testing the tools, will also have reached a level of maturity and ready to greatly expand the number of experts involved in this work so that many more Tool Development Groups could be set up to manage the development of an increasing number of new tools.
- The work of setting up the "Networked Local Digital Twins towards the CitiVerse" EDIC will likely have come to an end, and this would be a good time for a high-profile launch of its program of work.

If timing allows the formal launch of the "Networked Local Digital Twins towards the CitiVerse", EDIC would be an opportune moment for the start of scaling up activities. A key focus will be on high profile

communications and knowledge-sharing activities to raise awareness of the tools and the support systems in place, and to publicise all that had been learned about the practicalities and value of LDTs. The aim would be to support much larger numbers of communities moving forward with LDTs and starting to use the Toolbox.

Alongside the focus on scaling up, this would also be the time to utilise all the learning gained through the work and the various communities of experts to provide useful input to wider initiatives. Legislative regulations are foreseen to enter the market during this period, e.g., the AI Act is expected to be approved by the end of 2023, and probably to be enforced by 2026. A key task at this time would be to assess how AI-powered LDTs, and their tools can align with these developments. A similar role can be played in relationship to other key legislation, for instance the Data Act and the Data Governance Act.

In addition, there will be a wide-open opportunity to take the knowledge gained into international standardisation work. The EC is already influencing work within the International Telecommunication Union ((ITU), one of the three Word Trade Organisation recognised International Standards Development Organisations) in relation to the CitiVerse and the Minimal Interoperability Mechanisms (MIMs)[11]. The substantial understanding that will have been gained about the practicalities of implementing inter-operable LDTs will be welcomed within the global city Digital Twin standards community.

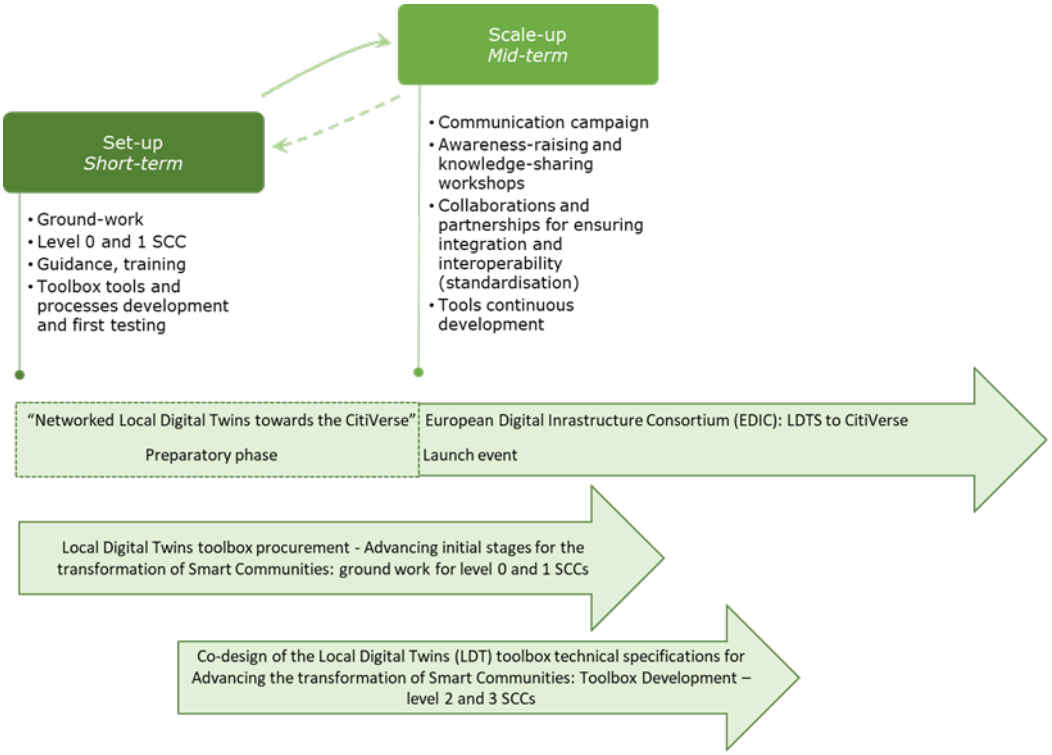


Figure 5: Mid-term scale-up

Source: Consortium

[11] https://mims.oascities.org/basics/oasc-mims-introduction

Long term – Sustain

Funding for the support for communities at the very beginning of their digital transformation journey through the "Advancing initial stages for the transformation of Smart Communities" project will come to an end in September 2025. The funding for the work of developing, deploying and evaluating the LDT Toolbox is expected to finish around a year later. An LDT Toolbox development work will, by that time, have taken on key aspects of supporting communities at the early stages of digital transformation as the scope of its work will have expanded to cover taking communities through all stages of LDT maturity.

However, all the initial support for the work of both projects will come to an end by the end of 2026. The challenge is that the work will have only just begun.

- There will still be many communities that will need help in starting on their digital transformation journey and beginning to implement and gain value from LDTs.
- There will be the need to develop many new tools to address new challenges as LDTs become more sophisticated.
- The aim is that the LDTs will become increasingly networked together so that all manner of linkages between cross sector and cross community Digital Twins will be made to address every more complex challenges.
- The role of the CitiVerse will also become increasingly important and many new tools will need to be developed to support all the Augmented Reality and Virtual Reality applications that will use LDTs to support applications centred around the citizen.
- Potentially, the ever-growing experience of developing LDTs and the CitiVerse in Europe will become of increasing interest globally and European companies will be in a prime position to address these new market opportunities.

By this time, the EDIC is expected to have taken on the overall responsibility for the roll out of Networked LDTs and the CitiVerse. However, the extent of the work needed will require extensive sources of revenue and support and sophisticated governance. Therefore, the long-term the focus needs to be on sustainability.

In preparing for the long term, the following factors need to be considered:

- sustainable governance;
- financial sustainability;
- technical sustainability;
- sustainable community; and
- sustainable scalability.

- To ensure that the Toolbox can be managed in the long term, the governance structure needs to remain stable. It is important that the governance will not be taken over by factional interests and remains in the hands of those with extensive management experience so that the organisation can continue to be agile and effective and meet the changing needs of communities. It is also crucial that a viable route to long term financial sustainability is identified, otherwise communities and industry may not have confidence in the long-term availability of the tools of the LDT Toolbox.

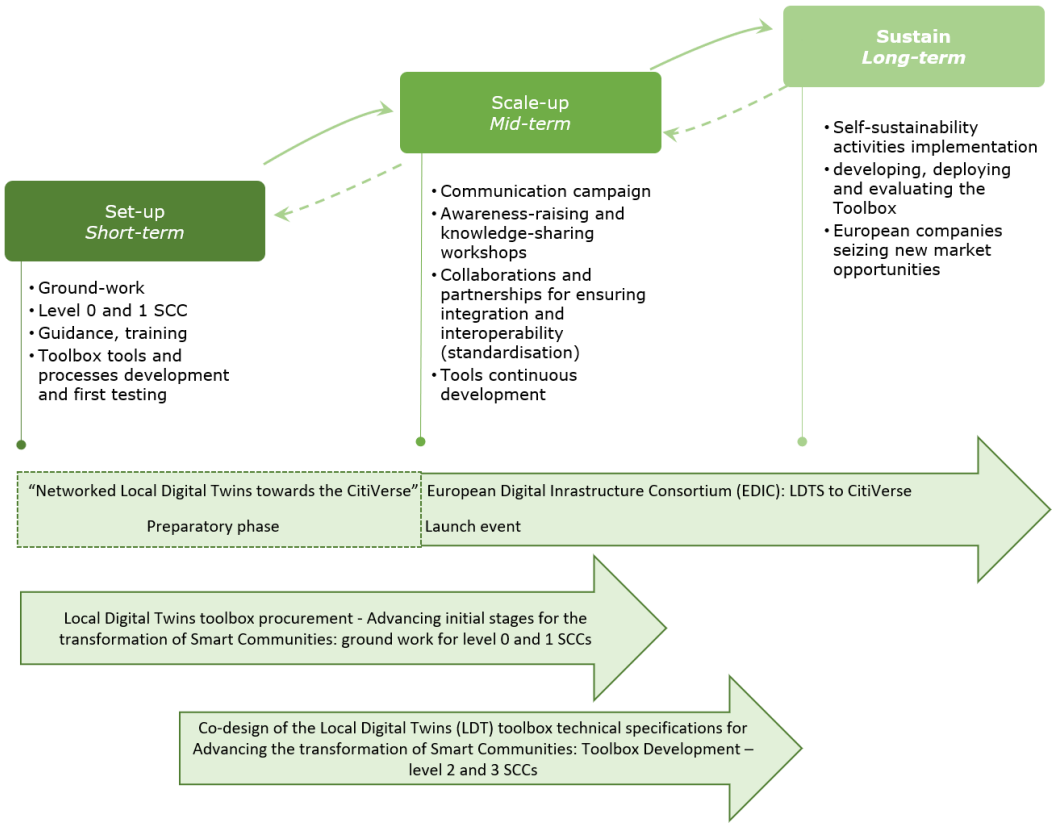


Figure 6: Long term – Sustain

Source: Consortium

Input for the roadmap

The description of these three phases is based on tenders in the LDT set of procurements and the time estimation is therefore subject to potential fluctuations. It nevertheless gives an overview of the interdependencies between the LDT Toolbox, and the evolution of the wider AI-driven landscape led by the EC. This timeframe was used to set the short- , mid-, and long-term of the roadmap.

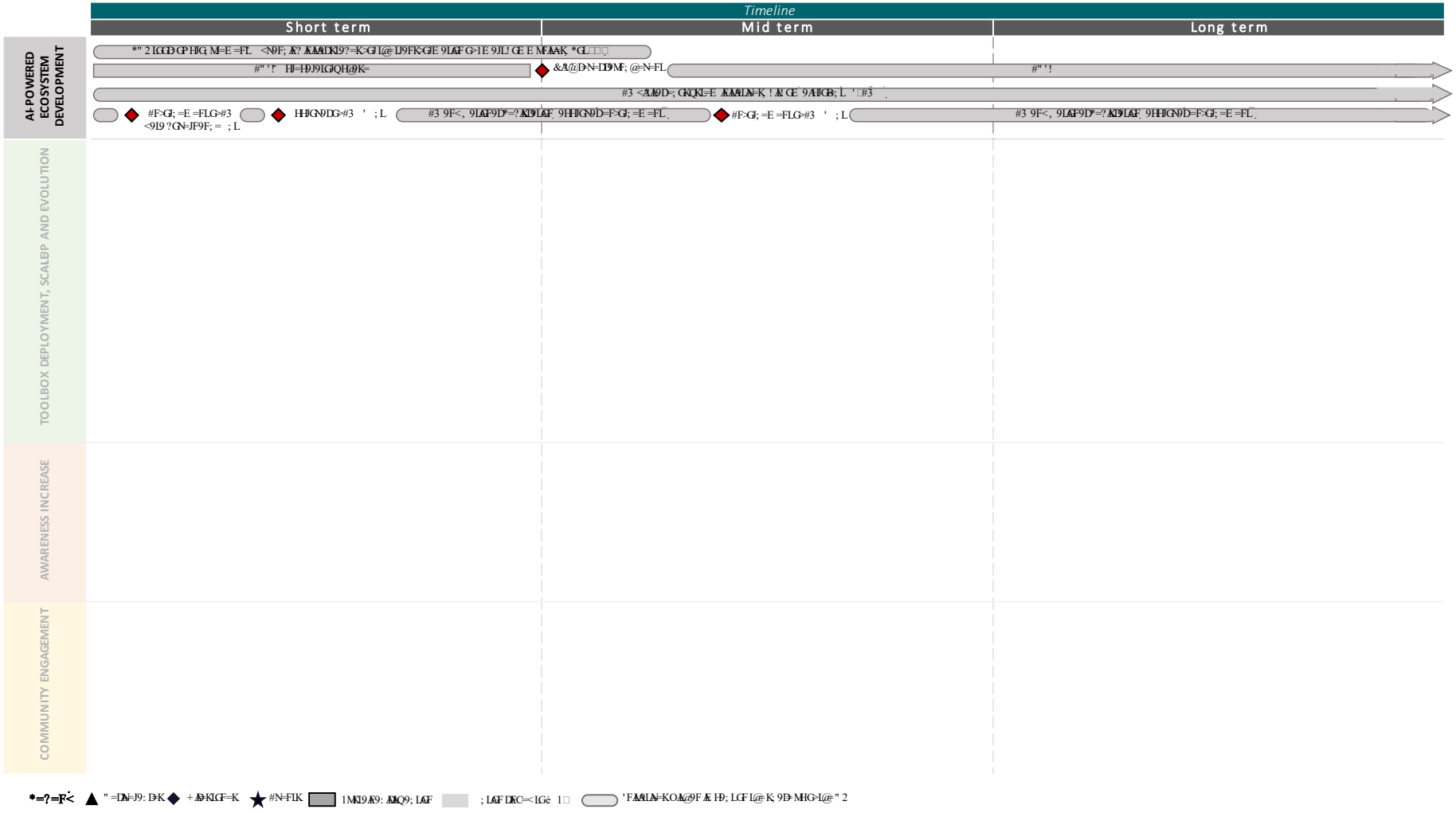


Figure 7: Inputs from 2.1 for LDT Toolbox geographical deployment, technical scale-up, and evolution roadmap

Source: Consortium

2.2 Initiatives in the EU Digital ecosystem

An LDT Toolbox's geographical deployment and technical scale-up and evolution can benefit from data spaces, Digital Twin initiatives and AI platforms. The EU digital ecosystem[12] outlines a diverse range of technology-focused initiatives and solutions. These are designed to enhance the implementation and use of digital solutions across EU cities and communities. This diversity is reflected in both, specific types of projects and broader organisations or networks. Together, they can be grouped into four distinct categories described below. Each category is illustrated by a selection from the nearly 100 technology-focused solutions[13] that exist within the EU's ecosystem. This is a first step in identifying the initiatives an LDT Toolbox could find synergies with to deploy and scale-up, and to be further analysed in an engagement plan.

Category 1: Promoting Data Sharing and Dissemination of Lessons Learned

The first category underscores the importance of data sharing and the dissemination of lessons learned. It is shaped by a combination of dedicated projects and collaborative organisations.

The **DS4SSCC project**[14] serves as a comprehensive guide in creating and managing data spaces. It promotes an open-source way of contributing, by fostering a collaborative environment to share knowledge on technical and non-technical BBs, necessary for efficient data interoperability, data trust and sovereignty, data value creation and data spaces governance.

CitCom.ai[15], a project launched in 2023 for five years, is another significant component of this ecosystem. If focuses on providing an AI testing and Experimentation Facility for Smart and Sustainable Cities and Communities. It is primarily meant for smart cities and communities to test AI solutions before they are fully integrated into city infrastructure. Data spaces and Digital Twins are used as fundamental vehicles to scale these AI solutions. CitCom.ai represents an excellent opportunity to test and experiment with LDT tools.

Furthermore, the **AI4EU project**, previously mentioned in Section 2.1, is a Europe-wide AI platform that offers tools, datasets and resources that promote the AI-enhanced sharing of data and lessons learned. It aligns with the objectives of promoting an open-source approach and fostering a collaborative environment to share knowledge on AI-driven data interoperability, trust, sovereignty, value creation, and governance.

Parallelly, the **Open & Agile Smart Cities** (OASC) organisation drives similar objectives but on a more organisational scale. By employing open data and encouraging citizen participation, OASC facilitates cities' ability to share their experiences and learn from one another. An essential outcome of the lessons learned by OASC member cities is the establishment of MIMs. These are tools designed to achieve interoperability of data, systems, and services between cities and suppliers around the world. These mechanisms provide a minimal common ground, allowing for diverse implementation but

[12] Second Cluster Meeting (06/06/23) on taking stock and re-visiting synergies and links between websites and interactive platforms, update on collective EU efforts on interoperability and standardisation (MIMs) and sharing of data and data sets to build up the Data Space for Smart and Sustainable Cities and Communities.

[13] See Annex 6.2 for a comprehensive list of nearly 100 technology focused solutions, relevant to cities and communities.

[14] Interactive Catalogue of Specifications for building data spaces in SCCs

[15] https://citcom.ai/ : Europe has recently been launching TEFs, Testing and Experimentation Facilities where complex digital technologies can be tested in real-world settings, physically and through simulation, from robots and AI to networking protocols and data processing and management. More information can be found here: https://digital-strategy.ec.europa.eu/en/faqs/testing-and-experimentation-facilities-tefs-questions-and-answers

ensuring interoperability across various technical architectures. MIMs enables seamless integration into any system or offering[16].

Lastly, the **Living-in.EU initiative** accentuates this approach further. The initiative focuses on encouraging European cities and communities to share best practices and provides a platform for discussions on digital transformations and solutions. This platform also serves as a centralised hub for sharing resources, which allows cities to assess their level of maturity and learn from each other's experiences.

Category 2: Creating Environments for Data Sharing and Collaboration

The second category emphasised the creation of an environment that enables data sharing or collection. The objective is to create an environment offering the necessary conditions for data ecosystems that can support a wide range of urban applications.

The DigiNEB.eu[17] initiative is at the forefront, focusing on creating a platform to host digital solutions and that is a valuable resource for city planners and stakeholders in the construction industry. Similarly, initiatives like Auroral[18], CommuniCity[19], DataVaults/PISTIS[20], DSSC[21], Matchup[22], and Scalable Cities[23], work towards the creation of digital platforms that promote data sharing and collection.

The AI orientation is strongly felt in DT4Regions[24] project, which aims to create a European Platform for Regions to share, connect, discuss and promote AI and Big Data solutions for digital transformation challenges in the Public Sector.

Furthermore, in this context, the Testing and Experimentation Facilities (TEFs) are important, as they create real environments where the latest AI-based software and hardware technologies can be tested in combination with other systems like data spaces. AI-powered Digital Twins are a type of umbrella term for different applications that use intelligence and data together to make decisions (e.g., on an operational or policy level). In this context the TEF for SCC citcom.ai[25] is not unique as TEFs are also used for agrifood, health and manufacturing. This is because an LDT is a system of systems that may need to connect with other AI-powered systems from these domains as well to tackle the societal challenges behind it. Thus, it is important to connect the TEFs and link them with the horizontal interoperability promises of the LDT reference architecture. Moreover, the upcoming regulations such as the AI and Data Act will impose different requirements on AI-powered Digital Twins in general. As is mentioned clearly on the TEF Q&A page[26] there is a clear relation between the TEFs and supporting the uptake of trustworthy AI in Europe, which is also a major concern for an LDT (see e.g., the Algorithm register).

Category 3: Data Collection, Analytics, and Actionable Insights

In this category, the emphasis is placed on the processes of data collection, analytics, and deriving actionable insights to inform decision-making.

[16] https://mims.oascities.org/basics/oasc-mims-introduction
[17] https://digineb.eu/
[18] https://www.auroral.eu/#/
[19] https://communicity-project.eu/
[20] https://www.datavaults.eu/about/
[21] https://dssc.eu/
[22] https://www.matchup-project.eu/
[23] https://smart-cities-marketplace.ec.europa.eu/scalable-cities
[24] https://dt4regions.eu/about-dt4regions
[25] https://citcom.ai
[26] https://digital-strategy.ec.europa.eu/en/faqs/testing-and-experimentation-facilities-tefs

Noteworthy projects such as CompAir[27], dRural[28], and Urbanage[29] are all examples of initiatives where data acquisition and analysis drive the digital transformation. By collecting, and assessing the data, these projects enable cities to gain insights, make informed decisions, and implement strategies for sustainable urban and rural development.

Another notable example in this category is Net Zero Cities, an initiative promoting zero-emission, clean, and sustainable cities. Part of the Horizon 2020 Research and Innovation Programme, it manages a repository of practical solutions, all aimed at helping cities to reduce their carbon emissions and move towards net-zero targets. This focus aligns with the EU's Green Deal, underlying the importance of digital tools in achieving sustainability objectives.

Additionally, this category encompasses certain initiatives under the Connecting Europe Facility (CEF)[30], such as Greenmov[31] and ODALA[32]. While Greenmov focuses on mobility data bolstered by AI-powered pilots, ODALA emphasises on the creation of Open Data Lakes. As these projects are currently coming to an end, the tools, standards, and open data sets they've developed can be used to boost the roadmap for LDTs. This emphasises the critical role of open data projects in facilitating the adoption and expansion of digital solutions in EU cities and communities.

Category 4: Focusing on Inclusiveness and Partnership

The fourth category emphasises the value of collaboration and building strong partnerships. It showcases how regional projects and platforms come together to amplify their impact. One of the main challenges for AI powered Digital Twin development at city level is their lack of or access to resources.

In countries like the Netherlands, Geonovum's[33] Digital Twins project demonstrates the power of regional collaboration. Similarly, in Sweden, the Digital Twin Cities Centre[34] stands out as an inspiration of joint effort. Both these initiatives promote the concept of "mutualisation", where cities or communities join forces to share resources and knowledge. This collaborative mindset is especially beneficial during the initial stages of setting up digital tools and systems. By coming together, cities and communities can leverage the combined strengths and expertise of multiple partners. For the LDT Toolbox, this presents an opportunity to serve groups of cities, enabling them to advance their digital ambitions, without making large individual investments. Instead, they can rely on the strength of partnerships to drive progress.

Common denominator

While the objectives of these four categories differ, they all converge on the themes of interoperability and standardisation, particularly highlighting MIMs(plus), and they uniformly recognise the diversity of topics and domains. These initiatives acknowledge that focusing on one specific topic is insufficient. Leveraging on the diversity of topics and domains, as well as the power of existing technologies such as AI, becomes paramount to drive innovation and address complex challenges in smart communities.

These initiatives highlight the importance of dialogue and incentives for cities to collect and share data, recognising that holding the data is not enough. It is the exploration and application of this data that

[27] https://www.compair-project.eu/
[28] https://drural.eu/
[29] https://www.urbanage.eu/
[30] https://cinea.ec.europa.eu/programmes/connecting-europe-facility_en
[31] https://green-mov.eu/
[32] https://odalaproject.eu/
[33] Digital Twins | Geonovum
[34] Digital Twin Cities Centre – A Vinnova competence centre (chalmers.se)

truly unlocks its value. That is where AI comes into play and enables cities and communities to identify patterns, predict outcomes, and make better informed decisions.

Ultimately, the current state of the ecosystem highlights how various projects and organisations share a collaborative spirit. Classified into three types based on their focus, they collaborate to promote data sharing, create environments for data collection, and drive AI-facilitated data assessment. Interoperability, domain diversity and AI-related technologies appear to be key pillars in driving innovation and facing challenges in Smart Cities.

Input for the roadmap

As we move forward with the roadmap, it is essential to continue leveraging and expanding these ecosystem initiatives. The roadmap should build on the existing collaborative foundations set by the various projects and organisations discussed earlier.

A strategic engagement plan needs to be drafted in the short term focusing on collaboration, data-driven decision-making, and AI-technology-focused solutions. By prioritising these, and in accordance with a successful implementation and wide use of the Toolbox data integration, standardisation and interoperability, we can make AI-driven LDT tools a cornerstone in EU Communities and Cities and set a standard for LDTs scale-up.

Figure 8 below, visually represents the key takeaways from this section, for the LDT Toolbox geographical deployment, technical scale-up, and evolution roadmap. As the report delves deeper, Section 2.3 will further explore some front-runner cases in the EU, to not only offer a blueprint of what is achievable, but also present the challenges and the strategies employed.

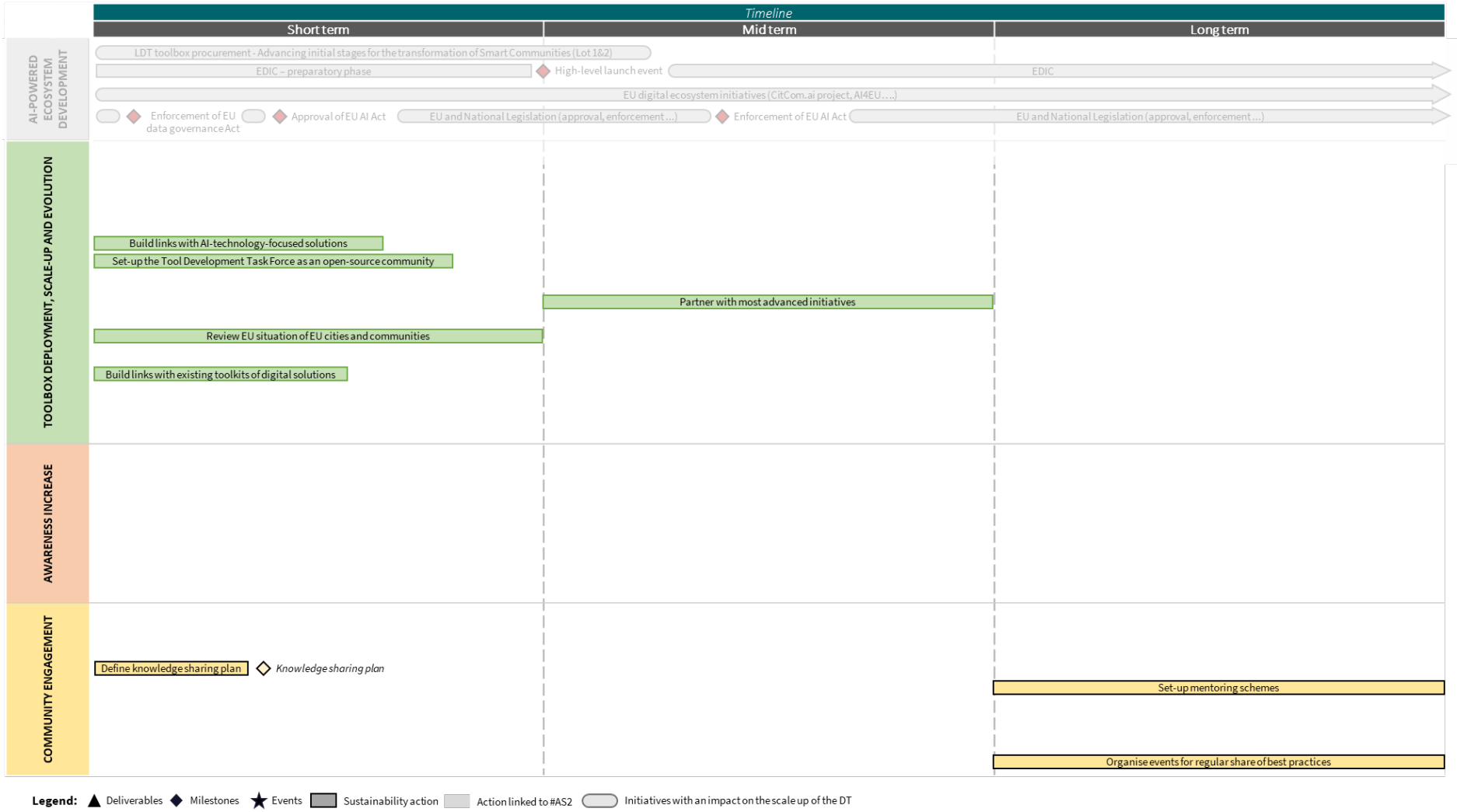


Figure 8: Inputs from 2.2 for the LDT Toolbox geographical deployment, technical scale-up, and evolution roadmap

Source: Consortium

2.3 Lessons learned from more advanced EU city data platforms

The synergies with the different initiatives and technology-driven solutions in the EU will foster the LDT Toolbox deployment and use by all types of maturity level communities. Lessons learned from more advanced EU city platforms are used to explore the idea of a City Guide on AI-powered Digital Twins.

As mentioned in the introduction, the analysis of the current state of LDT solutions implemented by cities and communities was based on the H2020 DUET project's Digital Twin Maturity Model. Out of the 135 LDT projects that have been identified and analysed in this project[35], 90% were found at the "Awareness" or "Experimental" stages, i.e., levels 1 and 2 of the maturity models. A minority (9%) was found at the "Predictive" (or level 3) maturity level. An LDT Toolbox, through its more advanced BBs, will allow current LDTs to evolve towards more intelligent maturity levels. However, from a practical point of view, implementing such advanced capabilities presents some challenges, particularly in terms of organisational readiness, capacity to manage complex systems, and the investment this type of innovative projects entails. Frontrunner cities such as Helsinki (Finland) and Rotterdam (the Netherlands) can offer some inspiration on how to address those effectively, and to build an actual community around LDTs (see Box 1 and Box 2 below). The examples below result from desk research in the context of this project and benefit as well from the case studies on digital innovation led as part of ESPON's DIGISER project. Both cities exhibit above average performance in digital innovation according to the DIGISER study[36], out of a sample of 248 European cities.

Box 1: Lessons learned from Helsinki's Digital Twin journey

Helsinki's Digital Twin

The development of Helsinki's Digital Twin has benefitted from a favourable digital innovation-friendly context created over the years in the city building on the following elements[37]:

- **Clear overarching vision(s) with boundable targets** (e.g., the "most functional city in the world by 2030, 2017-2021"[38]; or "Helsinki City Strategy 2021—2025: A Place of Growth[39]") and recognition of **digitalisation as an essential enabler** (e.g., Digitalisation program[40] and "the data generated by Helsinki is the most usable and used city data in the world by 2025"[41]).
- **Pioneering efforts to open public data from early on leading to new applications and services** (e.g., since 2011 the Helsinki Region Infoshare[42]consolidates data from the whole metropolitan region and incentivises its use by developers, researchers, students and businesses).
- **An ambitious Data Strategy[43] to develop the data, AI and robotisation capabilities of the City** to make better use of data and digital tools for proactive public service delivery, i.e., to anticipate needs of residents and visitors.
- **Strong experimentation culture embedded in the City, enabled by Forum Virium Helsinki**[44] (the City's independent innovation company) for prospective services which gives more flexibility to run experiments.

[35] For additional details on the mapping exercise including methodology, limitations, and key results, please check "D02.02 - Mapping EU-based LDT providers and users".
[36] DIGISER_CaseStudies.pdf (espon.eu)
[37] This is of course not a fully exhaustive list of the enabling conditions for digital innovation in the City of Helsinki. It builds mostly on the results of the Helsinki Case Study in the DIGISER project which combined desk research and anterviews to the City's CDO and other key ecosystem players.
[38] helsinki_city_strategy_leaflet.pdf
[39] Helsinki City Strategy | City of Helsinki
[40] Digitalisation program
[41] The world's best use of digitalization and best urban data | My Helsinki
[42] Helsinki Region Infoshare - Open data service (hri.fi)
[43] https://digi.hel.fi/english/helsinki-city-data-strategy/
[44] Frontpage - Forum Virium Helsinki

- **The City is a testbed for innovative solutions and opportunities for businesses** (e.g., Testbed Helsinki programme[45] for smart mobility, the Built Environment, etc).
- **Governance changes to address silos and promote the development of strategic digital innovation projects** (e.g., Digitalisation Unit, and the establishment of the role of the Chief Digital Officer).
- **Knowledge exchange and jointly pooling resources with other Finnish cities on projects of mutual strategic interest** including in the digital field (e.g., 6AIKA ("6Cities")[46] network with the six largest Finnish cities, namely Helsinki, Vantaa, Tampere, Oulu, Turku and Espoo).
- **Regularly engaging and driving the agenda in Global and EU initiatives and networks** (e.g., OASC's Personal Data Champion; MyData Global Network[47]; Cities for Digital Rights[48]; Living-in.eu).
- **Strong concerns with the design and impacts of the use of advanced technologies such as AI** (e.g., Helsinki AI Register[49]; signed the declaration of the Cities Coalition for Digital Rights).

The Digital Twin journey

Helsinki started its Digital Twin journey dates back to the 1980s[50]. The City has shown consistent commitment to this project over the years, which has been vital to its evolution, and which allowed the integration of new technologies and advancements over time (Figure 9). The City is considered[51] an **early adopter of computer-aided design (CAD), 3D city mapping, leading to full-scale Digital Twins**.



***Figure 9**: A timeline of Helsinki's Digital Twin evolution*

Source: Helsinki's pioneering city Digital Twin | VentureBeat

A relevant development took place in 2015 with the launch of Helsinki 3D+, which made use of new tools for capturing and organising a photorealistic 3D reality mesh and semantic data based on City Geography Markup Language (CityGML). The way Helsinki 3D+ is organised is presented in Figure 10.

[45] https://testbed.helsinki/en/
[46] https://6aika.fi/en/what-is-6aika/
[47] https://mydata.org/
[48] https://citiesfordigitalrights.org/
[49] Get to know AI Register – City of Helsinki AI Register
[50] Helsinki 3D | City of Helsinki
[51] Helsinki's pioneering city Digital Twin (venturebeat.com)

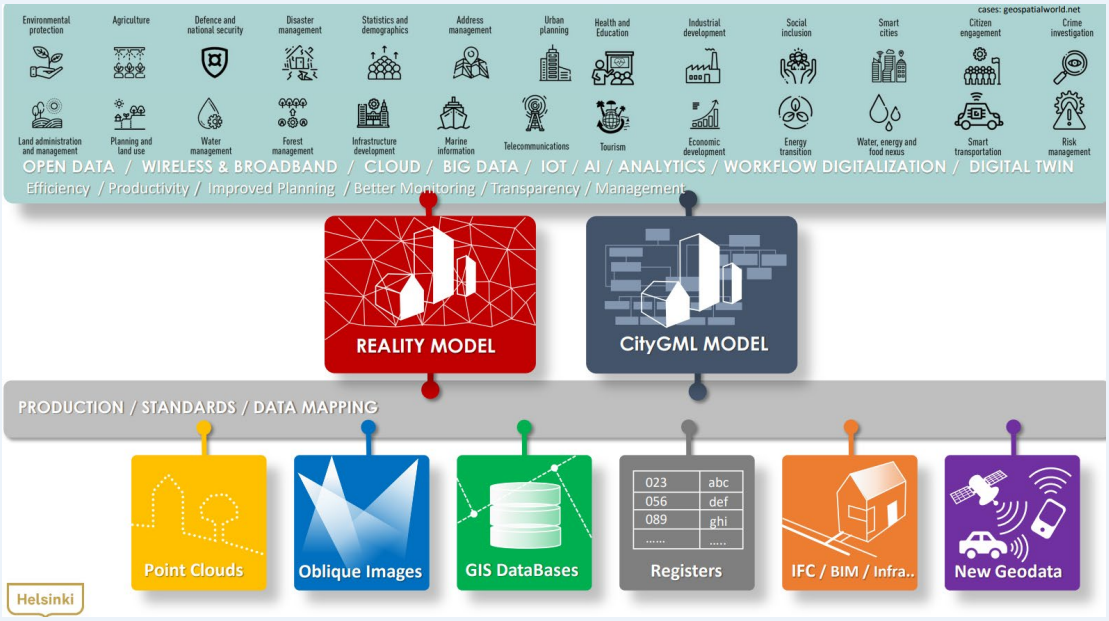


Figure 10: Helsinki 3D+ Digital Twins of a city

Source: PowerPoint-esitys (europa.eu)

From the 3D+ development experience, the City highlights the following challenges (quoting)[52]:

- Consistency between models
- Standardisation
- Data quality
- Data Interoperability
- Data maintenance / governance
- Going from "utopian" pilots to real-world use cases

Helsinki's Digital Twin encompasses a comprehensive dataset that includes both static information and near-real-time data. This diverse data integration has enabled a dynamic and up-to-date representation of the city. A key service in advancing the data integration efforts has been the Helsinki Region Infoshare service previously mentioned.

A key use case of Helsinki's Digital Twin is its **Energy and Climate Atlas.** This Atlas[53] provides comprehensive data on renewable energy and energy renovation opportunities in the Helsinki region. Through data modelling, it offers valuable insights into heating demands for buildings, changes with climate warming, and renewable energy prospects, such as estimates of solar and geothermal energy potential.

Another important DT milestone relates to the **Kalasatama Digital Twins project.** This was a pilot project in a specific city district that enabled experimenting with more detailed Digital Twin models of thar area relative to those of the overall city. For example, Kalasatama was exploring dynamic modelling possibilities of the natural environment towards what could be a potential Digital Twin of green infrastructure. The method of **Agile piloting developed in the smart district** was found successful as a testing platform in a real-life context.

Currently, Helsinki is focusing on enhancing the intelligence capabilities of its Digital Twin. This includes addressing the challenges associated with integrating advanced technologies and self-learning capabilities. In this respect, an important initiative of Helsinki's LDT strategy is the Helsinki AI Register, to provide transparency about the city's use of AI applications, their methods, the types of data they process, and the decision-making process they follow. In essence, it is a back-office platform that does not directly interact with the stakeholders but is accessible to the public.

[52] This overview comes from the presentation PowerPoint-esitys (europa.eu)

[53] For more information please consult the City of Helsinki's oficial website Helsinki 3D | City of Helsinki

Box 2: Lessons learned from Rotterdam's Digital Twin journey

Rotterdam's Digital Twin Journey

The pathway towards the **3D Digital Twin of Rotterdam** (Figure 11) has resulted from been successful thanks to the smart combination of different various factors such as:

- **The strategic importance of the Digital Twin as a tool to support the co-creation ambitions of the Digital City Programme** (DCP) of Rotterdam to stimulate co-creation. Citizen and stakeholder engagement as fundamental to build trust in the LDT solution.
- **The understanding that the essence of a digitally twin is to horizontally connect different silos** based on a common picture of reality. The City sees the digital twin not part of a smart solution, but rather as the basis for smart solutions to emerge.
- **A 'clear case for the LDT' through the strong belief in the City that the combination of the digital twin with other cutting-edge technologies**, like AI or the predictive power of big data, can contribute to significantly improving several city operations, from rescue operations by providing detailed information on the building occupancy to firefighters, to the monitoring of road traffic and the water in the canals to optimise the operations of opening and closing the city's bridges**.**
- **Experimentation towards a fully-fledged has been driven by public funding, especially through European R&I programmes**. Initial phases of the project, such as the Proof-of-concept and Prototype, were supported by funding and collaboration with different partners from two main H2020 projects, namely the Espresso project, and the RUGGEDISED project.
- **The development of a robust Open Urban Data Platform from which new applications and digital services can be derived** including the possibility for third party business models.
- **A mindset change, from "software-driven" towards data-driven, with open data standards and following the MIMs**, makes it a reference in the area of data management. The City is relying on several existent data collection infrastructures, such as sensor networks, to develop the infrastructure. The LDT aggregates all these sources of information and systems, together under the same structure.[54]
- **LDT is developed in compliance with the CityGML 2.0 format**. This model includes essential registrations, height data, and both spatial and non-spatial information, to accurately reflect Rotterdam's geometric reality.
- **The establishment of an innovative procurement process**, through a competitive dialogue to find private/external partners to collaborate in further developing the platform.

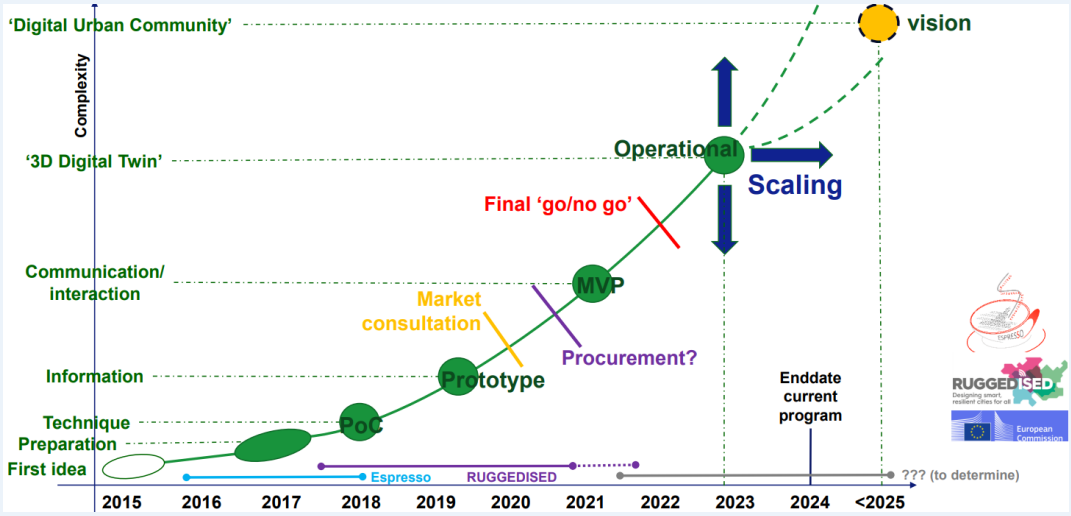


Figure 11: Rotterdam's roadmap of an open urban data platform and digital twin Source: PowerPoint-presentatie (oascities.org)

[54] Source: DIGISER_CaseStudies.pdf (espon.eu)

Input for the roadmap

The examples of Rotterdam and Helsinki show the importance of:

- Recognising **digitalisation as an enabler of the vision** or strategy of the city;
- Establishing a **smart governance structure[55] that coordinates internal efforts on digital innovations and enabling technologies** (digital twins, AI, Big data analytics, cloud computing,…);
- Aligning priorities and **pooling resources together** on common digital priorities with other cities and communities in the same metropolitan area or region[56];
- Encouraging an **internal innovation culture**;
- Leveraging **innovative procurement practices** that are more fit-for-purpose when it comes to riskier/more uncertain projects.
- **Agile piloting and experimentation** leveraging EU Research and Innovation programmes such as Horizon 2020;
- **Citizen-centric design** of digital public services and solutions;
- Leading significant efforts to promote **open data, data collection and data governance**;
- Promoting the **engagement with the innovation ecosystem**, including the possibility to create new services and business opportunities from data platforms and LDTs;
- Participating in EU projects and initiatives on the **fair and ethical use of AI**;
- Enabling agreements on the openness of the platform.

The roadmap should incorporate measures that enable cities to benefit from European funds effectively, while fostering collaborations among cities and other stakeholders, leveraging from existing and planned calls as well as relevant infrastructures being setup (e.g., SIMPL[57], and Data Spaces)[58]. This collaborative approach can facilitate sharing resources, expertise, and best practices. Developing a **city guide or checklist** for establishing an LDT, inspired by the examples of Helsinki and Rotterdam, can provide a practical and tangible path forward.

[55] An example is the creation of the role of the Chief Digital Officer (CDO) and digitalisation units.

[56] Examples: Helsinki (6AIKA; Innovation Ecosystem Agreements); Rotterdam (Metropolitan Region Rotterdam The Hague; G40 city network)

[57] Simpl: cloud-to-edge federations and data spaces made simple | Shaping Europe's digital future (europa.eu)

[58] Namely the Data Space for Smart and Sustainable Cities and Communities (DS4SSCC) and the Data Spaces Support Centre (DSSC).

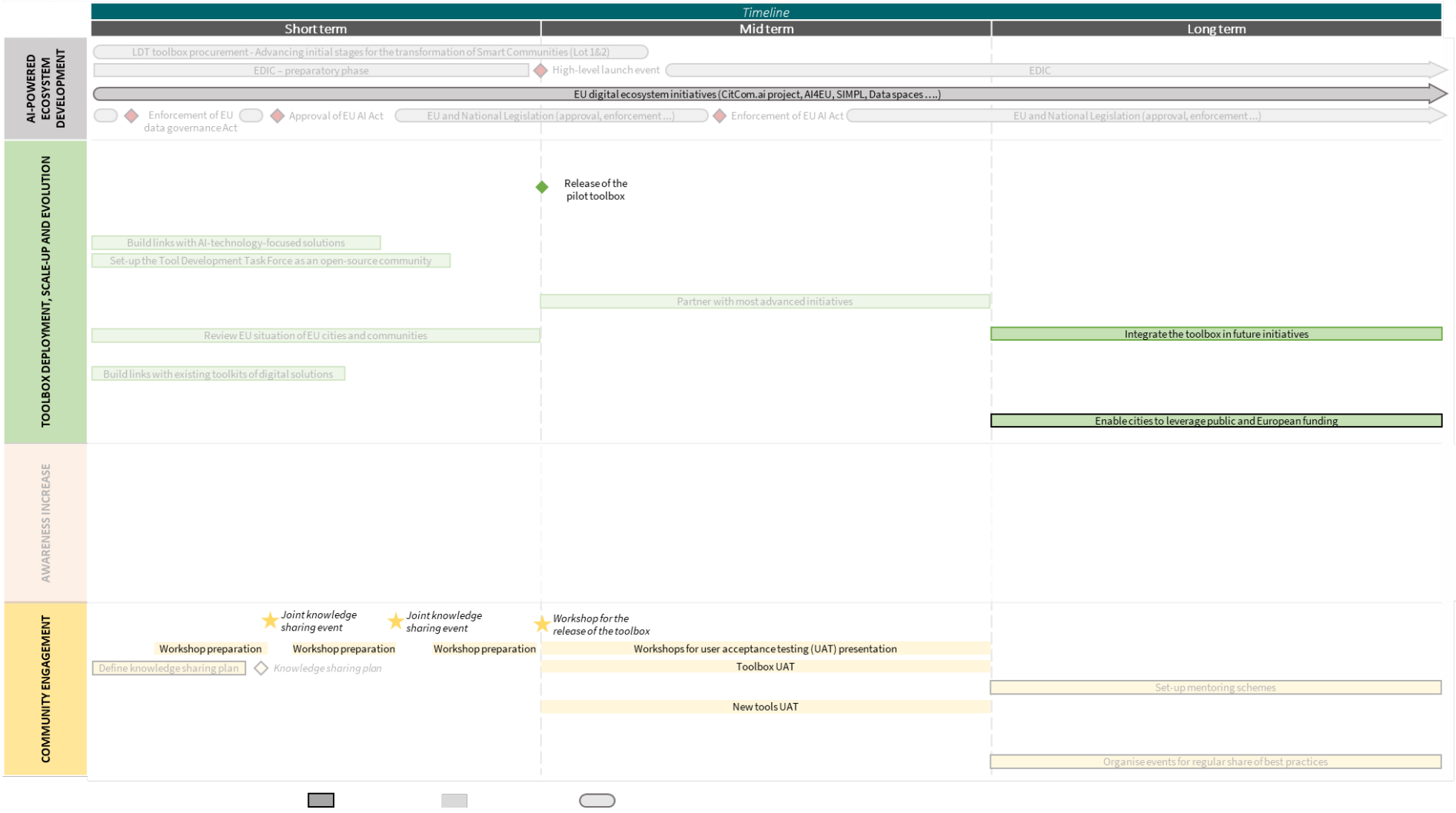


Figure 12: Inputs from 2.3 for the LDT Toolbox geographical deployment, technical scale-up, and evolution roadmap

Source: Consortium

2.4 Key challenges from Smart Cities and Communities

Smart Cities and Communities face a number of challenges when implementing LDTs. Identifying and understanding these issues is imperative for the roadmap's success, allowing for more robust and better-informed plans. According to the evidence gathered during the consultations, the key challenges have been depicted on the Figure 13 below.



Figure 13: Key challenges from Smart Cities and Communities

Source: Consortium

Input for the roadmap

To effectively navigate the challenges faced by smart cities and communities, a targeted communication campaign and dissemination actions are imperative. Such campaigns should primarily raise awareness about the significance of a robust data culture. The potential of LDTs in facilitating data-sharing, predictive analysis, as well as ensuring regulatory compliance should be highlighted. AI-driven analytics, like those found in the DigiNEB toolkit[59], can help to tailor these campaigns and raise awareness about data culture, for instance.

Addressing these challenges is not limited to communication alone. A "city guide", as suggested in Section 2.1, can serve as a practical tool for SCCs. This guide can delve into pressing issues such as cross-departmental collaboration, the lack of a governance model, financial constraints, etc. Moreover, this "city guide" could benefit from AI-powered insights, like those highlighted in the AI4EU[60] platform, which aims to foster the development and adoption of AI solutions in various sectors across Europe.

Another persistent challenge is the skills and expertise gap in smart city management. While bridging this gap is a long-term objective, Section 2.2 suggested a viable interim solution. By grouping several cities, communities, or stakeholders, there is an opportunity for enhanced knowledge transfer, resource-sharing, and identification of best practices. Such collaboration can serve as an inspiration for other cities.

Additionally, there is a strong need to achieve standardisation across different stakeholders and bodies. To ensure compatibility, interoperability, and consistency of systems and technologies in place, it is essential that different stakeholders and entities operate on unified standards. In addition, the data sources utilised need to be carefully registered. This not only helps traceability, but also facilitates the combination of standardised datasets, a cornerstone for the effective use of Digital Twins.

[59] https://digineb.eu/digital-toolkit?
[60] https://www.ai4europe.eu/

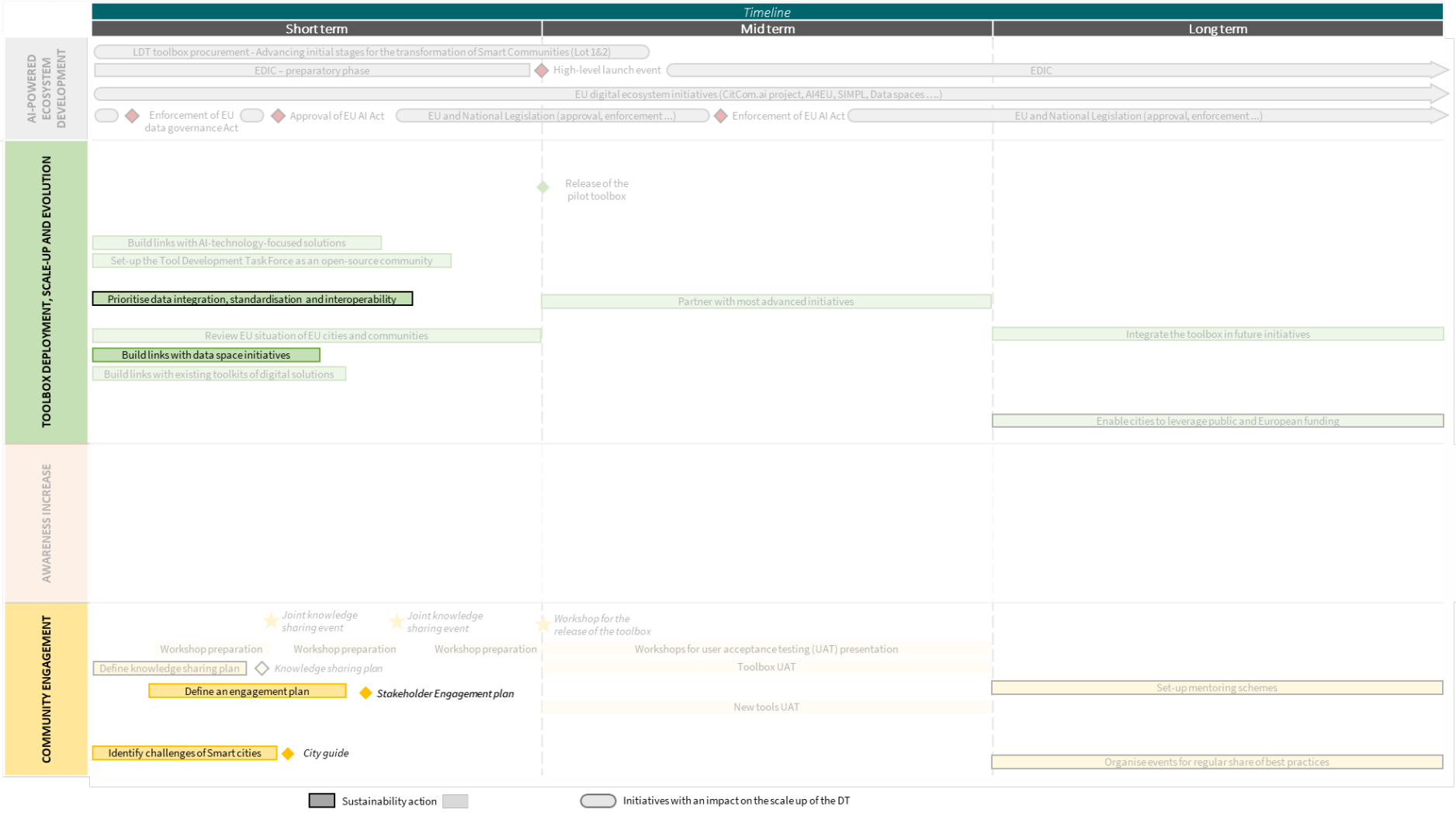


Figure 14: Inputs from 2.4 for the LDT Toolbox geographical deployment, technical scale-up, and evolution roadmap

Source: Consortium

3. Community Engagement

Building a community around LDTs not only involves the technical and development aspects of creating the LDT, but it also involves factors such as stakeholder engagement, community awareness campaigns, user training & support, and continuous feedback loops. While citizen participation and engagement are significant, it is important to remember that they represent just one dimension of a holistic community engagement approach. Evidence has shown that such active participation from citizens leads to an increased level of trust of institutions that supports integrated solutions for smart cities.[61]

A perfect illustration of the potential of stakeholder engagement and the power of visualisation and modelling capabilities of LDTs is the Luxembourg Institute[62]. The Institute demonstrated a cross-functional dashboard and a visualisation wall[63] to ministers and healthcare stakeholders to monitor the COVID pandemic, showing the impact of different scenarios to support the policy decisions. This is a good example of how a user-friendly interface, capable of visualising cross-domain data, can be used as an interactive playground. It facilitates the engagement of different stakeholders and enables the simulation of different scenarios[64] as demonstrated by several use cases in different countries.

Just as relevant here is the engagement with the administrations of local and regional communities for whom, and more importantly, with whom, the Toolbox will be built. Clearly, local community administrations are likelier to seriously consider tools that have been built in partnership with community administrations like them and not simply developed by technical experts. Tools built in this way are likelier to be appropriate to the community needs and to be implementable within the financial, regulatory and other constraints that local communities face. Because of this, a key factor in the success of the Toolbox will be how well it is able to engage with local communities, and through them, with the citizens who will be the ultimate beneficiaries.

The work will be greatly aided by the many existing organisations already active in engaging with those local and regional administrations, with whom collaboration will be essential for the geographical expansion, technical scaling-up, and evolution of the LDT Toolbox.

This section maps the main EU based organisations active in community engagement with which the project will be able to engage (3.1). The main community engagement actions are then described (3.2) against the three phases described in Section 2.1 – High level contextual timeline.

3.1 European Organisations and initiatives supporting networks of local communities

The work of engaging with the many communities around Europe to encourage them to see implementing LDTs and the CitiVerse as key aspects of their digital transformation journey will be greatly aided through collaboration with several key organisations and initiatives that support different networks of communities within Europe.

Living-in.EU supports over 140 communities that have signed up to the Living-in.EU declaration[65], supporting the European path to digital transformation, along with over 50 supporters from Industry and academia. It is proposed that Living-in.EU will set up and manage an Advisory Committee to guide

[61] The role of citizens in SCC solutions | Smart Cities Marketplace (europa.eu)
[62] LIST Demonstrates Collaborative COVID-19 Cross-Functional Dashboard to Ministers (chronicle.lu)
[63] Visualisation Wall | Luxembourg Institute of Science and Technology (list.lu)
[64] Microsoft Word - Urban Digital Twins v1.1.docx (imec-int.com)
[65] https://living-in.eu/declaration

in the setting up and management of the Community Organisation that will have oversight of the LDT Toolbox development and deployment.

In addition, there are several networks of local and regional administrations that can be drawn on for support, and these are listed below:

Eurocities – the network of the elected local and municipal governments of major European cities. The objective is to reinforce the important role that local governments should play in a multilevel governance structure.

OASC – is a non-profit, international smart city network, with its roots in Europe, that has the goal of creating and shaping the nascent global smart city data and services market.

CEMR – the Council of European Municipalities and Regions consists of 60 national associations of towns, municipalities, and regions from 41 of the 47 Council of Europe member countries, including the 27 Member States of the EU.

ICLEI – The European Secretariat of the Local Governments for Sustainability works closely with local governments and their partners to addresses local sustainability in Europe. They act as a facilitator between local government, and the private sector and has a strong focus on building capacity, sharing knowledge, and supporting local governments in the implementation of sustainable development at community level.

Major Cities of Europe is a European organisation of Local Government Chief Information Officers, Heads of Departments and Policy Makers. Its mission is to support local governments in sharing experiences, solutions and strategies in the use of information and communication technologies and leading-edge innovation in the administration of local governments and in the development of new services for the well-being and the wealth of their constituents and for the local economy. Membership is open to local government organisations, academic and research institutions and not-for-profit organisations and the private sector organisations can be involved through their sponsorship of specific activities.

Alongside this, as part of the work on the DEP-funded project on "Advancing initial stages for the transformation of smart communities"[66], research will be undertaken on smart city networks around the EU, for instance the **Spanish Network of Smart Cities** the **All-Ireland Smart City Forum**, and the **Smart Cities Network Austria.** Close contact will be made with each of them from the earliest stages of the development of the Toolbox and their support will be elicited to raise awareness of the value the Toolbox can provide.

The same DEP project will also set up a network of proximity partners at national level that will support the future development of a federated European Smart Communities Data Space, and a Local Digital Twins (LDT) network. This will enable the project to draw on the knowledge of those proximity partners in developing effective community engagements strategies within each Member State.

The Community engagement work will also be able to draw on the work of the European Digital Innovation Hubs (EDIHs). These are one-stop shops supporting companies and public sector organisations to respond to digital challenges and become more competitive. These are linked together in the EDIH Network, which is currently comprised of 151 EDIHs that are co-funded by the European Commission´s Digital Europe Programme and 76 EDIHs with a Seal of Excellence.

Finally, another set of communities to engage with will be the pilot projects on Data Space deployment. Work is underway on developing several sectoral data spaces across Europe. The Data Space for Smart and Sustainable Cities and communities (DS4SSCC) is particularly relevant, but others focus on areas

[66] See previous call for tenders https://etendering.ted.europa.eu/cft/cft-display.html?cftId=11876

such as Health, Industrial, Agriculture, Finance, Mobility, Green Deal, Energy, Public Administration, and Skills. Initially, the work has focused on building the tools and governance rules for each of these, but over the next few years there will be funding to support the implementation of data spaces across Europe. Many of these deployment actions will involve local and regional communities, who will clearly be interested in developing LDTs as part of the value to be gained from the data sharing ecosystems that they will be setting up.

3.2 Community Engagement over the Three Phases – Input for the roadmap

Throughout the three phases (short, medium and long-term), the key aspect of community engagement is the setting up and management of the Community Organisation and the Tool Development Task Force, where local communities and their representatives drive the strategy and are centrally involved in the development of the tools.

The Community Organisation is the strategic organisation that will drive the deployment of the Toolbox. It will be a cross-sector organisation, with membership consisting of communities, companies and other interested organisations, but with communities taking the major share of the leadership.

The Tool Development Task Force will be an open-source community to build and test the tools. The work will be done by small groups of communities working together with industry and academia to manage the development and testing of individual tools. These communities will be encouraged to set up working groups within each community to provide the opportunity to local SMEs and volunteer citizen experts to engage in the process. By this design, community engagement is built into the very heart of the building, development and testing of the tools and of the deployment strategy.

While the Community Organisation and Tool Development Task Force inherently infuse community engagement into their operations, there are additional strategies and methods of community engagement. These methods will evolve and adapt accordingly to the progression through the short, medium, and long-term phases.

In the short term

The work will be based on the organisations supporting networks of local communities that are mentioned above. In addition, the Living-in.EU mailing list serves as a database of key stakeholders and interested entities, enhancing the reach and connection to relevant audiences. Simultaneously, the stakeholder groups for the DS4SSCC and other data spaces can provide further sources of publicity and community engagement.

The need will be:

- To publish the support systems that will be put in place by the “Advancing initial stages for the transformation of Smart Communities” and drive use.
- To engage with cities or communities (i.e., those classified as LDT at level 2 and above[67]) to:
    - Learn from their experience and the tools they have developed or used
    - Invite them to join the Community Organisation and help guide the deployment of LDTs across Europe
    - Invite them to send representatives to the Tool Development Task Force and help with the development of the tools they need

[67] 1.1 Introduction: LDT level 1: Awareness of Twins, Level 2: Experimental Twins, Level 3: Predictive Twins, Level 4: Intelligent Twins

The engagement and publicity work directed at cities at a low level of digital transformation will be medium to high profile, focussing on publicising the support systems that are being put in place to raise awareness and interest in the Toolbox.

For work with communities that already have a level 2 or level 3 LDT, the engagement will be comparatively low-profile, but with a much deeper level of engagement. This is because it is important that those managing the Toolbox are not overwhelmed with numbers at this stage but rather that they can build strong relationships with those communities that are already committed to developing and using LDTs.

One idea might be to work with the organisations mentioned above to organise a joint knowledge sharing event. Communities that have already deployed LDTs would share their experiences on the role of LDTs in city management. Such event might consist of one or two days of knowledge sharing and then one or two days where representatives of other stakeholders are invited to join them in developing the detailed design of the Community Organisation that will be set up as part of the Toolbox Development and Deployment project to promote the roll out of LDTs across Europe.

There could be a subsequent event for technical representatives from local communities, with an initial one or two days of experience sharing and then work on setting up the Tool Development Task Force – the open-source community that will help to develop and trial the tools.

In the medium term

In addition to the organisations worked with already, the smart city networks identified by the research in phase 1 and the Proximity Partners in Member States will also be able to get involved. There will also be the opportunity to build on the links of those communities that have already become members of the Community Organisation.

The need will be:

- To continue to publicise the support systems that will be put in place by the “Advancing initial stages for the transformation of Smart Communities”.
- To publicise the LDT Toolbox to encourage its use.
- To publicise the Community Organisation and the Tool Development Task Force and encourage membership.

A particular focus, dependent on the eventual timing, is the likely launch of the EDIC. This will be an opportunity to bring together key community influencers in interactive workshops to enable the detail of the work of the EDIC to be co-created by national government representatives, community leaders and the EC and is an ideal time to meet and align with the broader Toolbox stakeholders. This is where Living-in.EU could have an effective organising role.

Overall, the focus in the medium term will be to further consolidate the Toolbox’s role within the wider European Smart City community and the related community and stakeholder support systems.

In the longer term

The same organisations involved in the medium term will continue to support community engagement in this phase as well. The challenge will be that those communities most open to developing LDTs are likely to have been reached during the previous phases and it may be a greater challenge to continue increase the numbers of communities using the tools. On the other hand, there will be growing evidence of the value of LDTs and increasing experience of cities across Europe of how to implement them.

Community engagement at this stage may increasingly focus on setting up mentoring schemes where communities with a mature deployment of LDTs share their experience with those who are yet to

begin. National and regional authorities will have an important role to bring clusters of communities together and support peer support and mentoring. In addition, other existing linkages between cities and communities may be of use, for instance *traditional* city twins, crossing inter-European borders and supporting digitally underdeveloped regions.

In addition to the ongoing challenge of reaching and engaging with an ever-growing number of communities, there will be a continual need to keep the work of the support systems and the Toolbox fresh and relevant, by keeping in touch with the developing needs and opportunities related to LDTs and the CitiVerse that are already part of the community.

Community engagement will consist of ongoing participation in events, including regular sharing of best practices across tool development, deployment, and management practices, and general online interaction via multiple channels to keep the individual local contributor as engaged as the advanced maturity level city representatives. There may well also be value in promoting awards schemes, both Europe-wide and in the individual member states to highlight both new and effective ways to develop and use LDTs, and also the role of community mentors. Therefore, the support of such community facilitation activities by the Toolbox organisations is essential, and an integral part of their core objectives.

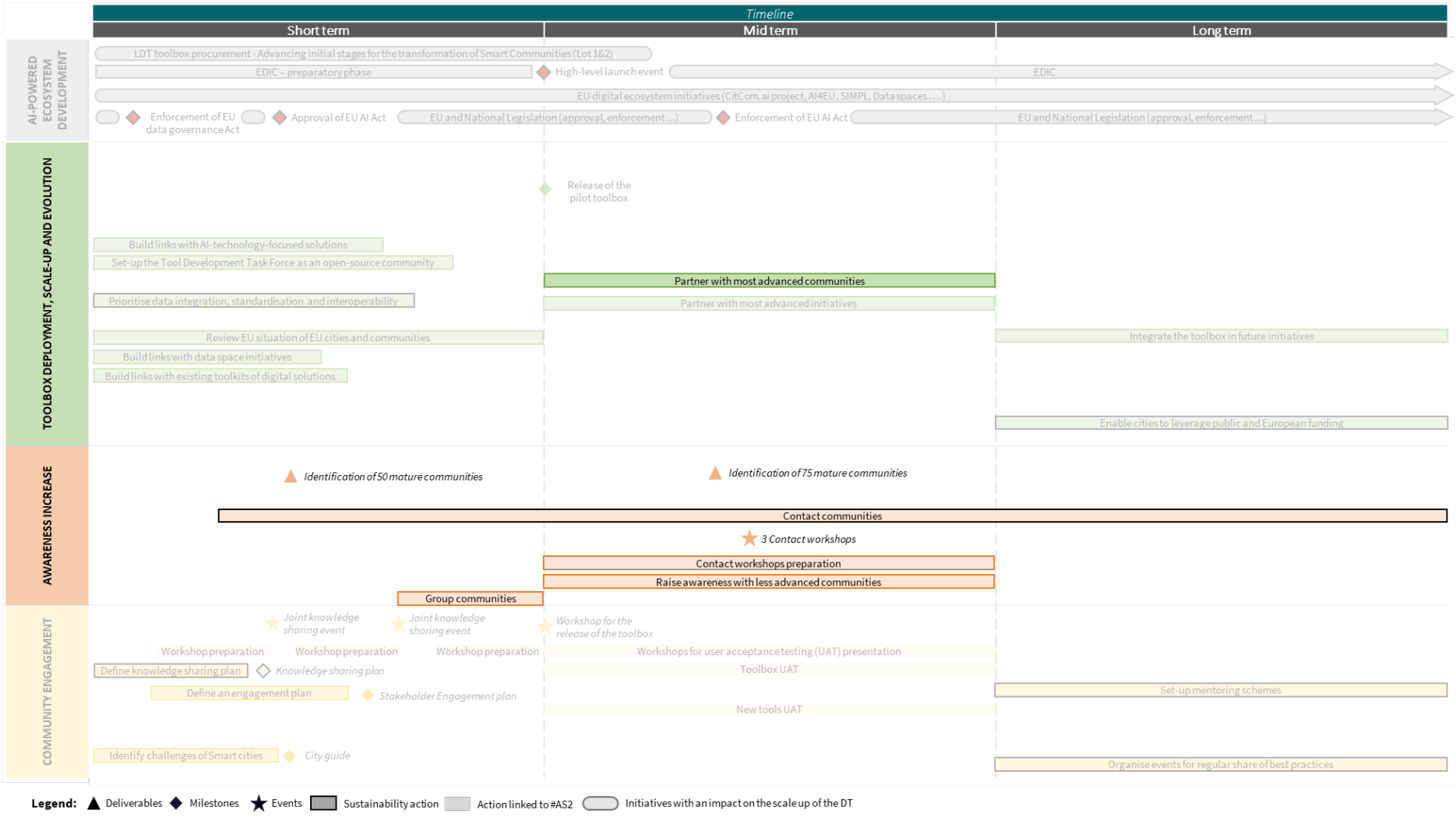


Figure 15: Inputs from 3.2 for the LDT Toolbox geographical deployment, technical scale-up, and evolution roadmap

Source: Consortium

4. Business sustainability plan - Maintenance and sustainable evolution of the LDT Toolbox

The business sustainability plan of an LDT Toolbox aims to control the impact of digital adoption and innovation, enhancing operational efficiency, staying adaptable to market trends, engaging with the community and ensuring transparent reporting. This plan aims to create a socially responsible, economically resilient, and environmentally conscious business that fosters innovation and builds stakeholders' trust.

In that sense, the maintenance of an LDT Toolbox and its components serves a crucial purpose of sustaining and extending the Toolbox service life, while mitigating security risks, service outages, and failures. Additionally, it enables an LDT Toolbox to adapt to new specifications and embrace functional and technical opportunities. It's important to view maintenance as a strategic process to sustain the value of the entire LDT ecosystem and to support its evolution.

The development of a business suitability plan for an LDT Toolbox has as main pillars, core sustainability principles defined for the LDT Toolbox – strategic sustainability, stakeholders ecosystem sustainability, financial sustainability, environmental sustainability, technological sustainability and legal sustainability.

Strategic Sustainability

This creates value for stakeholders and partners, establishing effective governance structures to guide sustainable decision-making, fostering innovation to develop relevant tools and BBs, ensuring scalability of sustainable practices throughout the Toolbox evolution, tracking specific targets related to sustainability goals and regularly evaluating results to maintain accountability and continuous improvement, ultimately aiming to achieve lasting positive effects on both the Toolbox and its wider ecosystem.

Stakeholders Ecosystem Sustainability

This focusses on establishing strategic partnerships, leveraging synergies, enhancing communication and transparency, fostering engagement within the community, and ensuring stakeholder retention. It is crucial to engage and collaborate with all stakeholders, including Toolbox partners, by understanding and addressing their needs and expectations. To secure stakeholders' buy-in, the Toolbox needs to effectively communicate the value and benefits of sustainable tools, aligning them with stakeholders' interests and concerns.

Financial Sustainability

This focuses on identifying a viable route to long-term financial sustainability, which is essential because without this the wider communities may lose confidence in the availability of the tools and Toolbox in the long run.

Environmental Sustainability

Over the business and technology landscape, leaders face increasing opportunities and responsibilities in achieving net-zero and environment sustainability objectives. In that sense, the LDT Toolbox should focus on minimizing the environmental footprint and impact of its infrastructure and technology tools by conserving resources and adopting eco-friendly practices. This pillar emphasises responsible resources management, sustainable sourcing, waste reduction, energy efficiency and efforts to mitigate climate change impacts.

Besides assessing the LDT Toolbox lifecycle impacts, by continuously monitoring relevant evaluation criteria, environmental sustainability requires putting a set of core initiatives in place:

- Implementing a holistic, integrated data and insights program to measure and drive environmental sustainability
- Leveraging a sustainability-driven tech strategy, and
- Driving transparency and accountability in the LDT Toolbox value chain.

Technological Sustainability

This embraces innovative technologies that can drive sustainable practices, allowing for the development/ implementation of new BBs and/or tools. To gain the support of communities in investing resources for implementing the tools, it is necessary to demonstrate how the technical development and maintenance work can continue in the long term. Development involves creating and enhancing the tools, while maintenance ensures their ongoing functionality, stability, and reliability. Good processes must be in place to keep all stakeholders engaged in collaborative work within the community.

Legal Sustainability

This adheres to laws, regulations and industry standards is a cornerstone of sustainability. Legal compliance ensures that an LDT Toolbox operates within ethical and legal boundaries, avoiding reputational damage and legal consequences while contributing to a fair and just society. By considering these principles and requirements and implementing a Business Suitability Plan to address them, the LDT Toolbox can be managed successfully in the long term, meeting the aspirations for enhancing the lives of citizens throughout Europe.

4.1 Maintenance and Evolution Value Model

The value model for the maintenance and evolution of an LDT Toolbox is based on inputs from the report 05.03 “Roadmap proposal for the deployment of the LDT Toolbox”, as well as on the main pillars for the development of the Business Sustainability Plan. Overall, the maintenance and evolution of the LDT Toolbox requires a proactive approach, continuous assessment, and a willingness to embrace innovation to stay ahead in the LDT landscape.

The maintenance of an LDT Toolbox will require a strong balance between ensuring the high availability of BBs and tools (BBs Utilisation) and fostering reduced maintenance costs (cost control). This way it will be possible to increase the service life of the LDT Toolbox, mitigating security risks, service outage and failures, while also being able to update the Toolbox over time. Figure 16 illustrates the proposed value model for a LDT Toolbox maintenance, which revolves around two key drivers for continuous improvement:

- Toolbox Building Blocks Utilisation; and
- Costs Control

This approach allows an understanding of where the potential is for value and therefore to adapt the LDT Toolbox utilisation accordingly to specific communities. Both the key drivers require the competences of security and service management, reliability engineering and continuous maintenance management. These act as key competences of the value model, being the link between the two drivers.

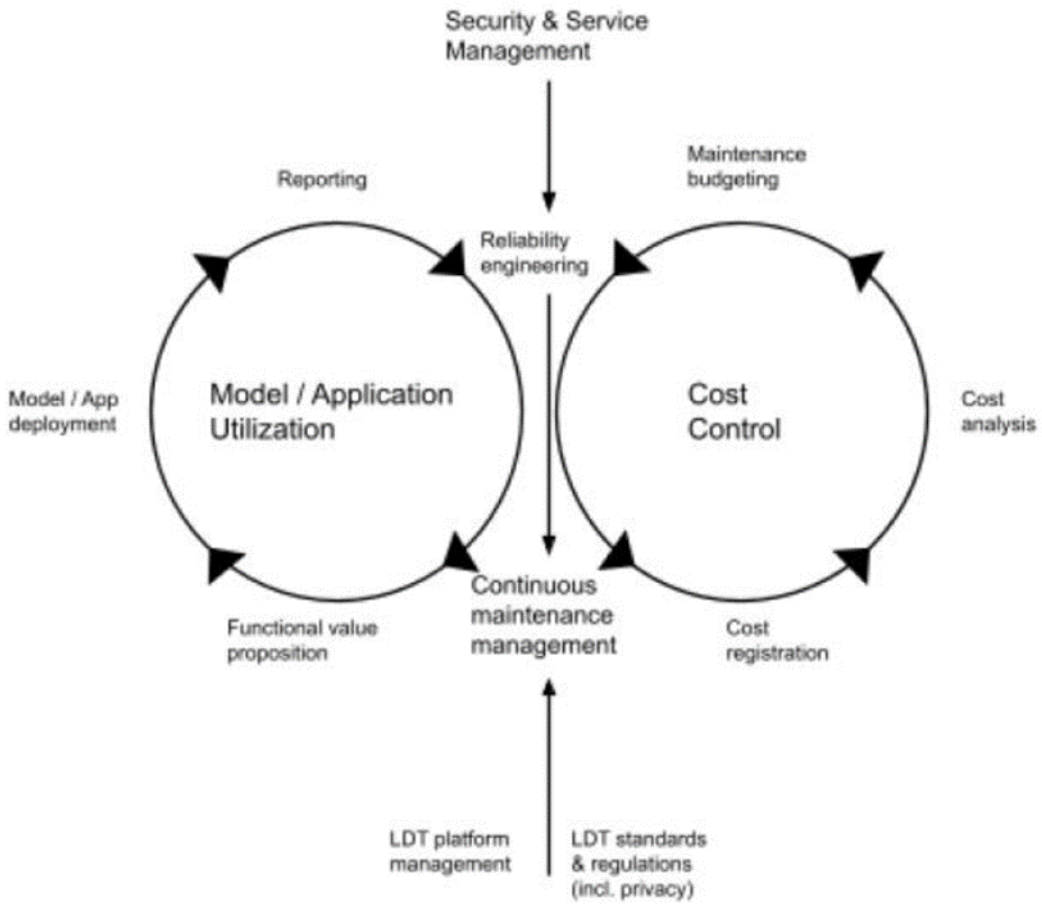


Figure 16: Value Model of the LDT Toolbox

Source: Ten CNECT/2022/OP/0098 – Technical offer

The impact of security and service management is significant on the value and costs of an LDT implementation, along with the overall management of the platform as a value chain and updates to standards and regulations related to LDT Toolbox exploitation. The key drivers of the value model of the LDT Toolbox follow a repeating process, as described below.

Building Block Utilisation

Functional Value Proposition: The Functional Value Proposition is a statement that outlines the unique benefits and value that the LDT Toolbox offers to communities or contributors based on its functionality. It highlights how the features and capabilities of the offering address specific needs or solve problems, setting it apart from competing alternatives.

Building Blocks Deployment: Building Blocks Deployment refers to the process of implementing and integrating modular tools or building blocks into the Toolbox. It involves integrating, configuring, and making use of individual building blocks to build a functional and scalable solution.

Reporting: Reporting involves the systematic gathering and presentation of information or data related to specific activities, events, or processes. It typically includes the generation of structured documents or visualizations to communicate findings, progress, or results to relevant stakeholders for decision-making and analysis.

Cost Control

Cost Registration: Cost registration involves creating a comprehensive budget that outlines expected expenses and revenues. It also includes strategic partnerships management to secure favourable terms and discounts, leveraging synergies and investing in emerging technologies. This activity sets the groundwork for effective cost control by establishing clear financial guidelines and optimising resource allocation.

Cost Analysis: Cost analysis entails a detailed assessment of expenses to identify Building Blocks or tools of inefficiency or overspending. This involves categorising costs as fixed or variable and employing efficiency improvements, to enhance productivity. Additionally, monitoring BBs or tools performance against industry standards help identify opportunities for improvement. Through continuous analysis, informed decisions on an LDT Toolbox can be made to optimise its cost structure.

Maintenance Budgeting: Maintenance budgeting involves regularly reviewing and adjusting budgets based on changing circumstances. It includes ongoing partners management and evaluation to ensure optimal terms, continued stakeholders training and incentives. By consistently maintaining and adjusting budgets, the LDT Toolbox can proactively manage costs and maintain financial stability.

The combination of key drivers allows for the execution of two further critical activities:

Continuous Maintenance Management: Continuous Maintenance Management is an ongoing and systematic approach to managing maintenance activities for assets, facilities, or equipment. It involves planning, scheduling, executing, and optimising maintenance tasks in a continuous manner to ensure the reliability, safety, and longevity of the assets while minimising downtime and maximising efficiency.

Reliability Engineering: Reliability engineering aims to ensure that BBs and/or tools consistently fulfil their intended functions while minimising instance of downtime, disruptions, and maintenance costs. It aligns with the broader goal of achieving value-driven BB and tools management by optimising the balance between performance, risk, and costs.

4.2. Monitoring Process to look for new technologies and methodologies

This section discusses a high-level description of the monitoring process required to keep the LDT Toolbox up-to-date in a sustainable way. The selection of BBs and tools for the Toolbox was initiated with a focus on ensuring its usability and adaptability to stakeholders needs. Nevertheless, it is crucial to recognise that the Toolbox must remain adaptable and dynamic, considering the ever-evolving technology landscape and changing stakeholders' expectations. In that sense, this section outlines the necessary activities to maintain the relevance and effectiveness of the LDT Toolbox over time. The possibility to integrate any new tools and technologies to be provided by the market or strategic partner ecosystem in the future should also be factored into the roadmap, as the LDT context is constantly evolving. This calls for a monitoring process based on the following main steps:

Step 1 Identification of new technologies

The identification of new technologies requires a strong engagement with existing stakeholders, market researchers and partners ecosystem, and a deep follow-up on overall standards and best practices (e.g., open-source practices), as well as AI-powered technologies. The identification process is iterative, requiring continuous monitoring and evaluation, along with systematic documentation and analysis of gathered information to remain responsive to evolving market dynamics and ensure continued relevance and value to stakeholders. Scanning and ecosystem analysis should therefore be a continuous process, through which it will be possible to ensure awareness of new technology trends, missing BBs or missing updates on existing BBs and/or tools.

Step 2 Assessment of new technologies

As the LDT Toolbox is dedicated to serve cities and communities across Europe, all the changes or evolutions to its tools or BBs, must be properly supported by data. This means that in addition to being aware of the latest technology developments and emerging stakeholders needs, it is also important to perform a more in-depth assessment when they seem to bring added value.

Thus, this step predicts a business case, including a feasibility study and a cost-benefit analysis, to be conducted, to decide and prioritise any new technologies, tools or BBs in the LDT Toolbox strategic roadmap and to identify any potential risks related to implementation.

Step 3 Identification of strategic partners

The mapping exercise of the LDT landscape in the EU under Task 2[68] shows that there is already a comprehensive set of technology providers and developers operating in the EU LDT market. This project identified 193 organisations, of which 159 are technology providers focusing on one or more Digital Twin capabilities[69], and 34 are technology providers who are not yet operational in LDTs but have the potential to enter the LDT domain as well. This step will ensure that there is a match between the new tools, technologies and/or BBs and the existing landscape of technology providers, so that these new tools can be timely added to the LDT Toolbox and adopted by cities and communities.

Finally, considering the several activities described above, sustainability plays a crucial role in LDT Toolbox effectiveness and relevance. By maintaining its dynamic nature through periodic updates and incorporating new technological advancements and stakeholders, the Toolbox will continue to provide valuable support to its stakeholders.

Input for the roadmap

As result of the maintenance and sustainable evolution section, to ensure the continuity and success of the LDT ecosystem, the following initiatives should be considered in the roadmap:

- **Develop a business sustainability plan** - This plan will serve as a guideline for the LDT Toolbox to ensure the fulfilment of core sustainability principles defined, while ensuring long-term success;
- **Define metrics to cost control** - This approach supports the avoidance of unnecessary expenditure, increasing the LDT value;
- **Define KPIs to monitor tools and BB utilisation** - This approach supports the optimisation of the tools and BB utilisation, decreasing costs and maximising the LDT value; and
- **Execute the monitoring process for new technologies and methodologies** – This process will ensure that the LDT Toolbox will always be aligned with emerging technologies and methodologies.
- **Involvement of the EDIC** – The long-term management and support of the Toolbox will be undertaken by the "Networked Local Digital Twins towards the CitiVerse" EDIC. All of the above plans, therefore, need to be drawn up in partnership with the working group setting up the EDIC so that they can be embedded in the proposals for how the Toolbox is developed, deployed and governed, once the EDIC takes up its role.

[68] See "D02.02: Public report on the mapping of the EU LDT landscape and its SWOT analysis including details of the dissemination session for more details".

[69] Capabilities were defined according to the Digital Twin Capabilities Periodic Table made available by the Digital Twin Consortium.

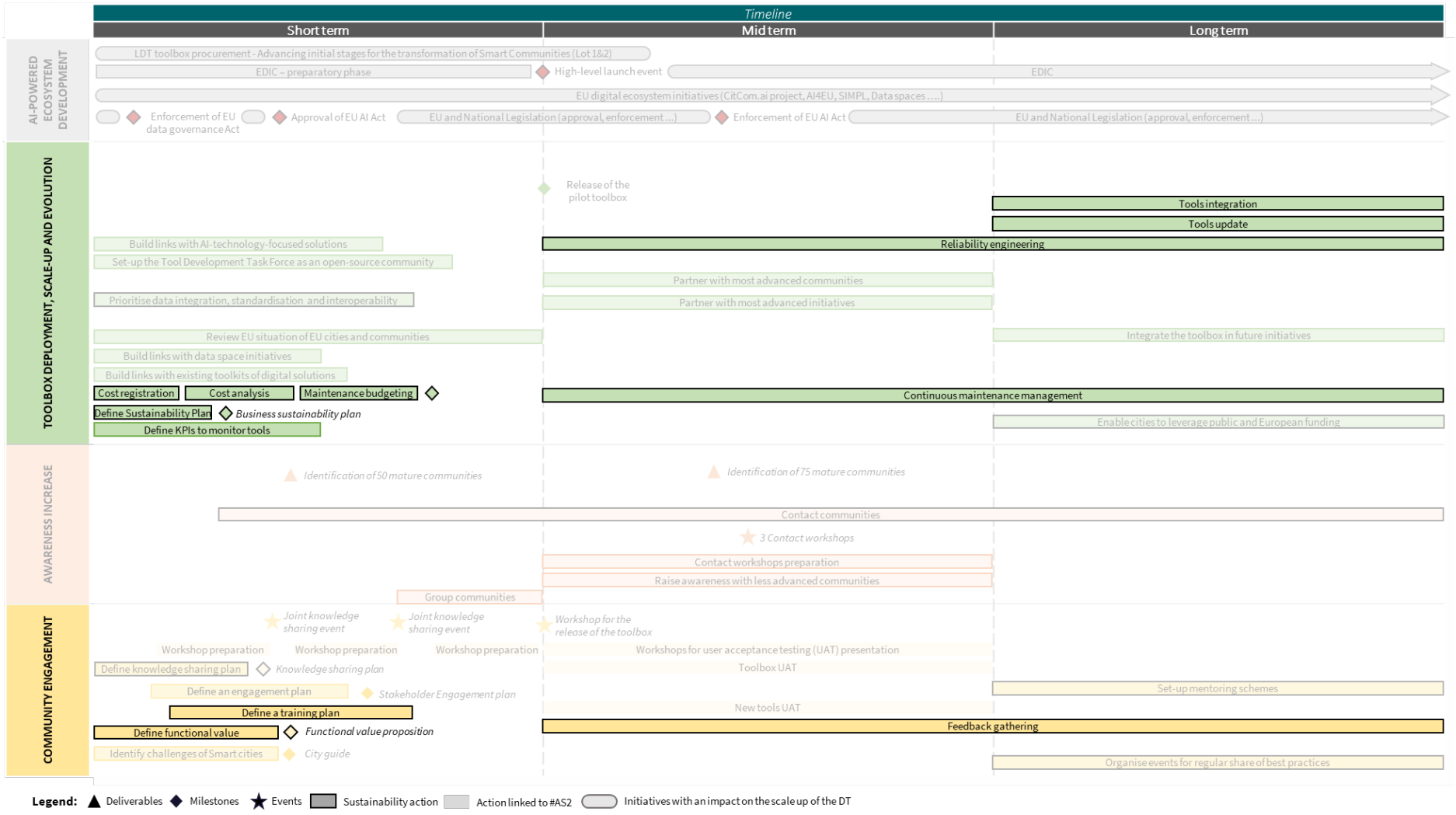


Figure 17: Inputs from 4.2 for the LDT Toolbox geographical deployment, technical scale-up, and evolution roadmap

Source: Consortium

5. High-level roadmap for the LDT Toolbox geographical deployment, technical scale-up, and evolution

This section will include a definition of the major milestones of the roadmap for the geographical deployment, technical scale-up, and evolution of the LDT Toolbox towards the scale-up of AI-powered Digital Twins, based on assumptions (5.1) and guiding principles (5.2) concluded from the previous sections 2, 3 and 4. The risks and mitigation solutions of key actions (5.3) will then be developed before strengthening the roadmap with the definition of KPIs for the assessment and progress monitoring (5.4).

5.1 Assumptions

An LDT Toolbox geographical deployment and technical scale-up, and evolution roadmap is underpinned by a set of assumptions related to the previous reports of this project, and the analysis on the eco-system conducted in Section 2, including the challenges identified during the consultation phase.

Table 1 : Assumptions for the LDT Toolbox geographical deployment, technical scale-up, and evolution roadmap

#	Assumptions
AS-1	The roadmap is articulated on short-, medium-, and long-term sequencing. These are based on the Section 2.1 high level contextual timeline but the phases per se, and their approximate timing, are not considered so as not to invalidate the roadmap in case of any changes. *See Section 2*
AS-2	The roadmap on the LDT Toolbox both geographical deployment and technical scale-up and evolution in relation with the EU AI-powered Digital Twin development is building up on the LDT Toolbox high-level deployment roadmap as developed in D05.03 "Public report on the roadmap proposal for the deployment of the LDT Toolbox", discussed and validated by stakeholders during the project's interim workshop on 12 July 2023.[70]
AS-3	The LDT Toolbox both geographical deployment and technical scale-up and evolution will leverage on other data spaces and Digital Twin initiatives as well as communication actions. *See Section 2 - Link to R2*
AS-4	Interoperability (both horizontal and vertical), integration, user experience, and re-usability are the most important features to build on the content of the LDT Toolbox for its geographical deployment and technical scale-up and evolution. *See Section 2 and D02.02 conclusions*
AS-5	The technical scaling-up for the LDT Toolbox roadmap will target the countries exhibiting significant involvement and market engagement, showcasing a relatively strong foothold in the LDT landscape. *See D02.02 conclusions*
AS-6	The geographical deployment and the awareness-raising, knowledge-sharing actions of the LDT Toolbox roadmap will target peripheral European countries noticeably lagging in terms of representation of LDT providers. *See D02.02 conclusions*

[70] The list of participants also included representatives from the organisations part of the Living-in.eu Steering Board.

AS-7	The geographical deployment, especially the vertical one, will focus on local and regional scale, including regional reach beyond countries boundaries encompassing additional regions or country groups such as Benelux. *See D02.02 conclusions*
AS-8	Stakeholders will be available to contribute to the actions. *See Section 2 and 3, and the roadmap actions*

5.2. Guiding principles

Based on the overview of the EU digital and AI-powered Digital Twin ecosystem (Section 2), the Community Engagement (Section 3), and the Business sustainability plan of the Toolbox (Section 4), a set of guiding principles were defined to pilot decision-making throughout the different phases of the LDT Toolbox geographical deployment and technical scaling-up and evolution roadmap. These principles are listed in Table 2 below.

Table 2: Guiding principles for the LDT Toolbox geographical deployment, technical scale-up, and evolution roadmap

#	Guiding Principles	Description
GP-1	**Activity Based Timeline**: Short-, medium-, and long-term planning	Based on the estimated high level contractual timeline (2.1), the choice is made to not mention any dates for the roadmap itself as there is a high risk of fluctuation. Rather, the roadmap for deployment and scale of the LDT Toolbox will be based on short, medium and long-term type of activities.
GP-2	**Interoperability and Diversity of Topics**	Interoperability and the diversity of topics and domains are a common denominator. These initiatives acknowledge that focusing on one specific topic is not sufficient, but that is rather essential to leverage the diversity of topics and domains, as well as existing technology, to drive innovation and address complex challenges in smart communities.
GP-3	**Collaboration and Data-driven Focus**	Emphasis on collaboration, data-driven decision-making, and technology-focused solutions to lever as many digital ecosystem initiatives.
GP-4	**Efficient Data Handling & User-centric Design**	The successful adoption of the LDT is driven by efficient data handling and user-centric design principles: most or all the actions of the roadmap must contribute to this objective.
GP-5	**Continuous User Engagement**	Need to remain continuously close to the users. This implies to engage with the community, understand their needs and challenges, and ensure the LDT Toolbox is continuously updated and improved.
GP-6	**Optimisation of Public and European Funding**	Maximisation of the use of public and particularly European funding.

5.3. Overview of the main streams of the roadmap

As mentioned in previous sections of this document, key inputs related to the wider EU digital and AI-powered Digital Twin ecosystem, community engagement, and the business sustainability plan of the LDT Toolbox informed the design of the roadmap. This roadmap is composed of four streams representing the multiple co-dependent activities which should occur simultaneously over the short-, mid-, and long-term so that the LDT Toolbox's geographical deployment, technical scale-up, and evolution can be positioned within the context of the wider deployment of AI-enabled LDTs, and ecosystem of ongoing and planned EU-funded initiatives and projects. As described in the high-level timeline taking into account the other tenders in the LDT set of procurements presented in section 2.1, "short term", "mid-term" and "long term" refer respectively to the "set up and implementation", "scaling up" and "sustainability" phases of the deployment of AI enabled LDTs. Figure 18 presents a consolidated view of all the roadmap visuals illustrating the previous sections of this document.

The **AI-powered ecosystem development** is the first stream which provides an overview of the various activities and projects that will affect the EU digital and AI-powered Digital Twin ecosystem as a whole and that will therefore influence the geographical deployment, technical scale-up and evolution of the LDT Toolbox.

The **LDT Toolbox deployment, scale-up and evolution** is the second stream which include the activities which will need to occur so that the LDT Toolbox's geographical deployment can take place (vertically and horizontally) and so that its scale-up and evolution will meet the needs of EU cities and communities characterised by all levels of maturity.

The **Awareness increase** is the third stream which refers to all the activities which will need to occur to raise awareness and interest in the Toolbox. This includes multiple workshops such as the city guide workshop which aims to raise awareness on the development of a guide for cities to set up a LDT and its related activities.

The **Community Engagement** is the fourth and final stream which include the activities required to engage with EU cities and communities not only to regularly share best practices in the development of AI-powered LDTs but also, to gather user feedback and opinions on the LDT Toolbox and be able to ensure continuous improvement, user engagement and satisfaction.

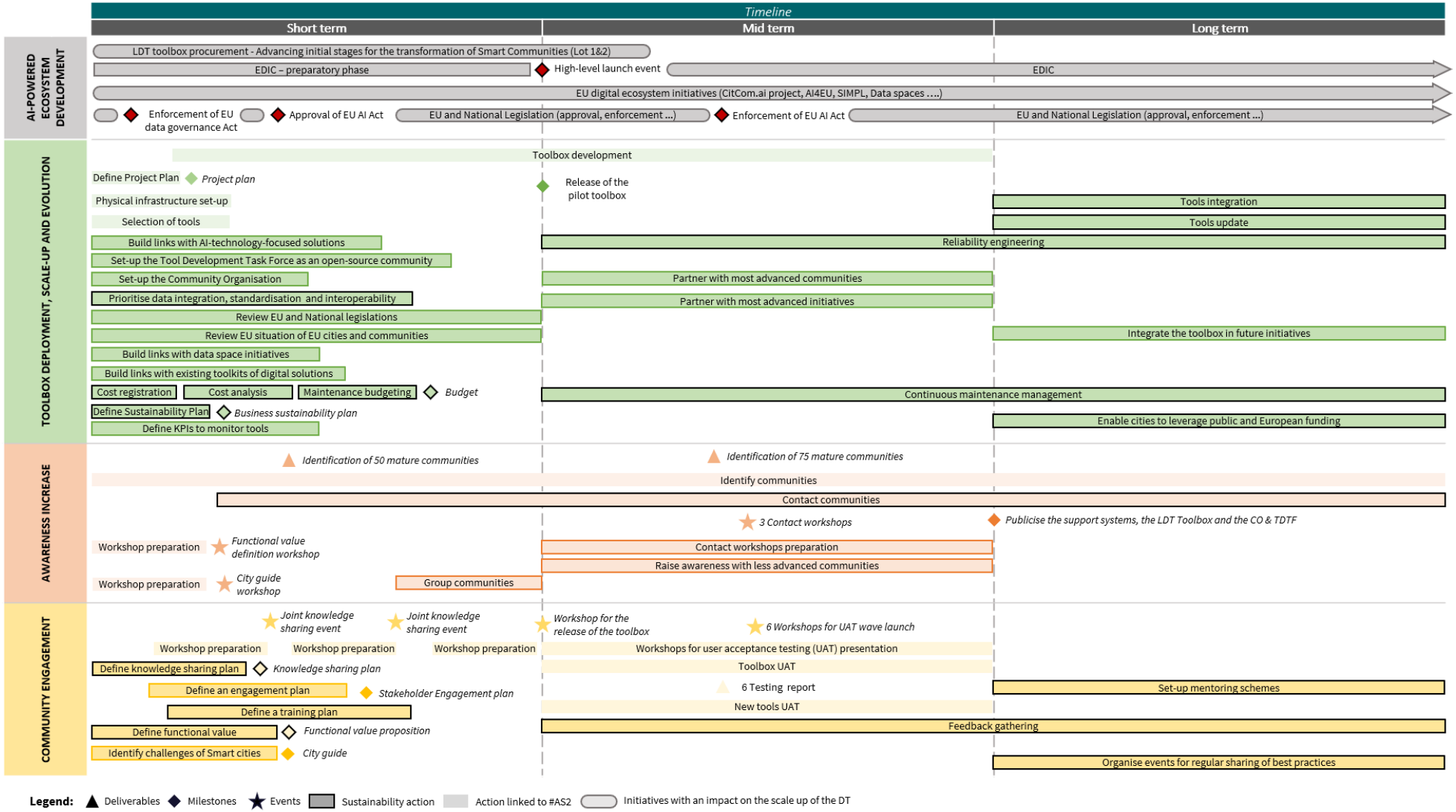


Figure 18: LDT Toolbox geographical deployment, technical scale-up, and evolution roadmap

Source: Consortium

5.4 Risks and mitigation actions

The risks and associated mitigation actions related to the LDT Toolbox are listed in Table 3:

Table 3: Risks and mitigation for the LDT Toolbox geographical deployment, technical scale-up, and evolution roadmap

#	Risks	Mitigation
R-1	Cities have different digital maturity levels. They might not see the value of a Digital Twin, or they see it as something very futuristic for them.	Inform cities about the value of local Digital Twins, and how this can bring value for Cities at different digitalisation levels. A BB is developed with guidelines on how to build a LDT Roadmap and a user-friendly guide is also suggested as a long-term action for the geographical deployment. *Link to AS-9*
R-2	Wider ecosystem is not static – the "LDT sector" is growing and new LDT platforms ought to be developed and implemented in the upcoming years.	The roadmap and course of actions need to be regularly updated based on a constant – or very active – monitoring of the "LDT sector". *Link to AS-2*
R-3	LDT Toolbox to be mainly deployed and scaled-up in already advanced areas.	Conscious focus on awareness-raising and knowledge sharing actions. *Link to AS-6*
R-4	Uncertainty, and non-linear behaviours can present a risk for a Digital Twin.	Calibration and validation of an AI model should be considered.
R-5	Ethical concerns over data ownership and privacy could result in low public trust Indeed, public trust can be compromised by cybersecurity risks that could include unauthorised access or modifications to the real data.	Security checks should start at the onset of a Digital Twin and cybersecurity training be developed.
R-6	Technical issues can concern data quality and accuracy that might reduce the public confidence when Digital Twins are used for decision-making. Indeed, if data presents low quality and accuracy, the predictive analysis could be misleading.	Quality checks and regular technical and data updates.
R-7	Uncertainty of obtaining funding, and risk of insufficient funding, which could hamper the digital transformation journey.	Ensure a proper financial planning, supported by a comprehensive business case and diversity of funding options.
R-8	Poor data management.	The implementation of a robust data management process, which specifically facilitates the development of a city guide or checklist for the setup of an LDT. *Link to AS- 9*

5.5. Definition of KPIs for assessment & progress monitoring

The objective of the KPI Lifecycle Management is to ensure the efficient deployment, adoption and scale-up of the LDT Toolbox across European communities. To evaluate the progress of each phase and the overall success of the deployment, adoption and scale-up, a set of assessment criteria must be determined. These reflect the project's objectives and expected outcomes (establish and roll out a strategic plan for the widespread adoption and expansion of AI-powered LDTs across EU communities and achieving LDT maturity across the EU) and can be evaluated based on quality (observing the adoption rate of the Toolbox and user satisfaction levels), speed (monitoring the pace of deployment and adoption) and financials (considering the overall project cost).

These KPIs are selected, evaluated, and allocated as follows:

- **On-time delivery**: compares the actual and planned completion times.
- **On-budget delivery**: compares actual costs with the planned budget, considering one-off and recurring costs variances.
- **Adoption and utilisation**: measures the number of communities using the Digital Twin solution and the geographic spread of its adoption. Furthermore, the adoption of the AI-powered LDTs in less technologically advanced communities could be tracked to assess the democratisation of this technology across the EU.
- **Data accuracy and integrity**: tracks data errors, discrepancies, and data validation rates to ensure reliable and trustworthy information for decision-making.
- **Satisfaction and Feedback:** compares the actual user satisfaction levels and the frequency of feedback gathering among different user groups throughout the deployment and scale-up stages.
- **Stability and Performance**: tracks system uptime, response times, and availability.
- **Post-deployment support:** measures the quality and responsiveness of the support services offered, including issue resolution time, support ticket response times, and customer satisfaction with support services.

Each phase of the project requires specific KPIs that should reflect the objective and outcomes expected to be achieved within each phase. These might consist of:

- **Short-term phase:** Since the end goal of this stage is a pilot/MVP, this involves tracking the time taken to develop the Digital Twin solution, to develop the AI models, the cost of development, and the number of development cycles completed, the amount of resources consumed in model development and training, and the accuracy of the models.
- **Mid-term phase:** this tracks the speed of deployment, the number of communities adopting the solution, the geographic range of adoption, and the quality and integration of the data being fed into the AI models. Additionally, tracking the 'training and onboarding speed' might be useful to consider, to ensure a smooth adoption of the Toolbox by the communities.
- **Long-term phase:** this measures the rate at which the Digital Twin solution is adopted by new communities, the geographic spread of its adoption, the growth of data sources successfully integrated into the system, the level of automation achieved, the model prediction accuracy, and the user satisfaction.

After defining these KPIs, it is crucial to set specific targets for each phase. This provides a clear benchmark for what constitutes success at each stage and ensures that we have clear and measurable goals against which we can evaluate our progress. Targets could be set-up according to the sustainability pillars defined in Section 3 of this report. This could be done as follows:

- **Strategic Sustainability:** this pillar refers to the ability to create value for stakeholders and guide sustainable decision-making. Tracking how the LDT Toolbox is effectively meeting stakeholders needs and innovating to developing relevant tools, could be done by, for instance, measuring a 'stakeholder satisfaction score' or the 'number of new tools developed'.
- **Stakeholders Ecosystem Sustainability**: this pillar refers to fostering engagement within the community and ensuring stakeholder retention. A 'User retention rate' or 'community engagement rate' could be used as KPI and a target could be associate to it.
- **Financial Sustainability:** this pillar refers to the economic resilience of the LDT Toolbox. Return on Investment or Total cost of ownership could be relevant KPIs to set targets to.
- **Environmental Sustainability:** this pillar refers to minimizing the environmental footprint of the LDT Toolbox. Looking at the 'Waste reduction rate' or 'energy efficiency' could help to gauge the environment effectiveness of the LDT Toolbox.
- **Technological Sustainability:** this pillar refers to the adoption of innovative technologies to drive sustainable practices. 'Adoption rate of new technologies' or 'system uptime' can assess the extent to which the LDT Toolbox is successfully implementing and maintaining new technologies.
- **Legal Sustainability:** this pillar refers to the adherence to laws, regulations, and industry standards. To ensure that the LDT Toolbox is operating within ethical and legal boundaries, the 'Compliance rate' or 'Number of legal issues' could be KPIs to set specific targets to.

Regular reporting schedules are established to continuously evaluate the KPIs, track progress, identify bottleneck, and keep stakeholders updated. This allows for necessary adjustments and improvements.

Finally, the next steps toward effective monitoring of the projects include Performance Reporting and Control and Review and Adjustment of the KPI model. This approach ensures that the project can be effectively monitored and guided towards successful completion, ensuring it meets its objectives on time, within budget, and with high user satisfaction and community engagement.

6. Conclusion

This report, provided at the completion of the project on scoping out work on the LDT Toolbox, presents a comprehensive analysis of the journey towards the implementation and scale-up of AI-powered LDTs within EU cities and communities. It provides a framework of the EU's journey towards digital transformation, powered by AI.

While the foundation was laid by earlier reports, emphasising the importance of community requirements, European values, and stakeholder involvement, this comprehensive document dives deeper into the geographical deployment, technical scale-up and evolution of the LDT Toolbox.

An important aspect of this report is its emphasis on collaboration. Drawing from surveys, interviews, and workshops conducted as part of this project, it is clear that the evolution of the LDT Toolbox will significantly benefit from partnerships with communities with already fairly mature LDTs in place. By collaborating with these experienced communities, the Toolbox can leverage the tools they have established and benefit from their lessons learned. This ensures that as newer communities embark on their digital evolution, the Toolbox will offer precisely the instruments needed, that will help them move through every stage of maturity.

The overview of the EU digital ecosystem, supported by a high-level contextual timeline, demonstrates the relevance and future potential of the LDT Toolbox within an AI-driven landscape. The success stories and challenges identified from Smart Cities and Communities provide insights into what difficulties might need to be addressed in the future. The LDT Toolbox emerges as a comprehensive reference framework, set-up with re-usable BBs from other EU initiatives. It underscores the significance of sharing knowledge, offering technical solutions for the continued development of LDTs, and prioritising continuous updates on new AI technologies and adapting to new solution assessments.

Moreover, by following up on the actual usage of BBs, gathering feedback from the community, and setting the tone for establishing necessary standardisation, the LDT Toolbox contributes to vendor-agnostic solutions based upon interoperability standards. The "Community Engagement" description emphasises the role of all stakeholders, ensuring the LDT Toolbox fits seamlessly into the larger ecosystem.

The business sustainability plan, proposed in the report, encompasses a maintenance and evolution value model, along with a monitoring process for identifying new technologies and methodologies. The constant scanning of the AI and technological landscape for solutions that could enhance the effectiveness and utility of the LDT Toolbox should ensure its long-term viability.

Furthermore, a high-level roadmap for the geographical deployment, technical scale-up and evolution of the LDT Toolbox with regards to the EU AI-powered DT has been proposed. This roadmap should serve as the blueprint for the successful deployment of EU AI-powered Digital Twins.

To conclude, the entire project provides a comprehensive guide for the design, development, deployment, and scale-up of the LDT Toolbox. This ensures that the LDT Toolbox is not only technologically advanced, considering the latest AI developments, but also aligned with the needs and expectations of the communities, as well as with the wider evolution and efforts towards a Single European data market. Embedded in a commitment to user-centricity and community collaboration, the Toolbox stands as a pillar of digital progress, set to adapt and evolve with the evolving AI-landscape and the diverse needs of tomorrow's European Digital Ecosystem.

Annex 1 - Listing of other initiatives ecosystem

Here are just a few toolkit type repositories that may be relevant channels to help publicise the eventual LDT Toolkit:

- DigiNeb - https://digineb.eu/digital-toolkit
- Net Zero cities (Home - NetZeroCities) – Initiative promoting zero-emission, clean, and sustainable cities – part of the Horizon 2020 Research and Innovation Programme
- Climate neutral cities mission (Climate-neutral and smart cities (europa.eu))
- Scalable cities (Scalable Cities | Smart Cities Marketplace (europa.eu))
- EIF4SCC (Proposal for a European Interoperability Framework for Smart Cities and Communities (EIF4SCC) | Shaping Europe's digital future (europa.eu))
- ICC (https://www.intelligentcitieschallenge.eu/)
- Community of practices (https://knowledge4policy.ec.europa.eu/territorial/community-of-practice-cities_en)
- EUI (https://www.urban-initiative.eu/about-innovative-actions)
- Covenant of Mayors (https://eu-mayors.ec.europa.eu/en/home)
- CIVITAS (https://civitas.eu/)
- Urban Data Platform Plus (https://urban.jrc.ec.europa.eu/?lng=en)
- COMPAIR (https://www.wecompair.eu/)
- iCapital (https://eic.ec.europa.eu/eic-prizes/european-capital-innovation-awards)
- Urbanage (https://www.urbanage.eu/)
- Rural digitalization: Auroral (AURORAL), dRural (https://drural.eu/)
- NEB New European Bauhaus: beautiful, sustainable, together. (europa.eu): DigiNEB
- Smart Cities marketplace (Creating smart cities together | Smart Cities Marketplace (europa.eu))
- Big buyers' platform (Big Buyers Working Together | Public Buyers Community (europa.eu))
- Living-in.EU (Join us in building the European way of Digital Transformation for 300 million Europeans | Living in EU (living-in.eu))
- EDIHs (European Digital Innovation Hubs | Shaping Europe's digital future (europa.eu))
- TEFs (Sectorial AI Testing and Experimentation Facilities under the Digital Europe Programme | Shaping Europe's digital future (europa.eu))
- Big Data test infrastructure (Big Data Test Infrastructure (BDTI) (europa.eu))
- DS4SSCC (DS4SSCC)
- Tourism DS (Data Space for Tourism - City Destinations Alliance)
- Destination Earth (Destination Earth | Shaping Europe's digital future (europa.eu))
- Knowledge Centre on Earth Observation (KCEO) (https://knowledge4policy.ec.europa.eu/earthobservation_en)
- Mobility DS (Mobility Data Space - Mobility Data Space – Data Sharing Community (mobility-dataspace.eu))
- Green deal DS (Green Deal Data Space GREAT Project The GREAT project)
- Built environment platform (METABUILDING, innovation Funding and Support for SMEs of the Built Environment Sector - H2020 - INNOSUP-01-2018-2020 (cluster facilitated projects for new industrial value chains) (metabuilding-project.eu))
- EU DSs (A European Strategy for data | Shaping Europe's digital future (europa.eu))
- Data Spaces support centre (Data Spaces Support Centre (dssc.eu))
- SIMPL (Simpl: cloud-to-edge federations and data spaces made simple | Shaping Europe's digital future (europa.eu))

- Intelligent cities challenge - Tech4Good marketplace
  https://marketplace.intelligentcitieschallenge.eu/en

ISBN: 978-92-68-06529-7



Publications Office
of the European Union